

Command Processor Operation

SW11-607

**Implementation
Startup and Configuration**

Command Processor Operation

**SW11-607
Release 682**

10/2011

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1. Introduction

1.1 About This Document

Purpose

Provides procedures for using the TotalPlant Solution (TPS) Command Processor functions, used in conjunction with virtually all Engineering Personality activities.

Intended audience

Anyone using the Command Processor to modify Default Volume Pathnames and perform file-managing utility commands.

For product release

TPS Release 682

Document usage

New users of the Command Processor should use this section to get started.

This document includes information on:

- Modifying Default Volume Pathnames
- Command Processor Commands
- Recovering from Errors

Additional sections of this document provide detailed descriptions of the commands.

Document conventions

The following standard document conventions may be used in this document. Check section 1 for conventions unique to this document.

Item	Convention	Meaning
<KEYCAPS>	Pointed brackets around a term	A key on a computer keyboard
[TARGET]	Square brackets around a term	For touch screen monitors, an area that can be touched to invoke a system action
entry data	Courier bold text	Data you enter into the system
screen/program data	Courier text	Screen or program data

1.2 Overview

Purpose of this section

This section provides basic information for users of the Command Processor functions. These functions include:

- Use of the Command Processor display to execute utility commands.
- Use of the Modify Default Volume Pathnames display to change default pathnames.

Topics in this section

1.3 Command Processor Display and Support Utility Menu

1.4 Uses for Command Processor Commands

1.5 Pathnames, Volumes, Directories, and Files

1.6 Modify Default Volume Pathnames Display

How do I know when to use the Command Processor?

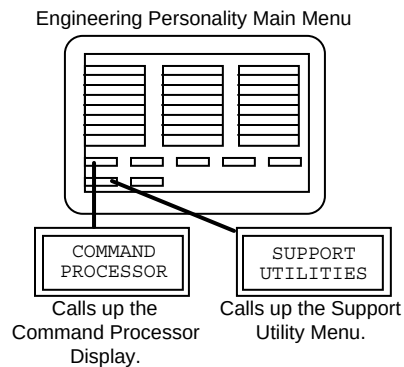
The documentation for Engineering Personality activities advises you when you need to use the Command Processor. Briefly, the Command Processor is used in the following situations:

- **System startup**—the Command Processor is used at specific steps in the startup of a TPS system. Procedures for the initial startup are provided in the *System Startup Guide*.
- **Data-entity building**— the Command Processor is used to establish and manipulate the volumes, directories, and files used in data-entity building. Entity building procedures are in the *Data Entity Builder Manual*.
- **Installing CL Programs**— the Command Processor is used in installing and modifying CL structures, and to display and print information relative to those activities. The procedures are in *Control Language/AM Data Entry Manual*.
- **System Engineering**— the Command Processor is used in almost all of the activities that are selected from the Engineering Main Menu. The *Engineer's Reference Manual* provides guidelines and recommendations that affect the design and implementation of your system, including several recommendations for the user of the utilities in such work.

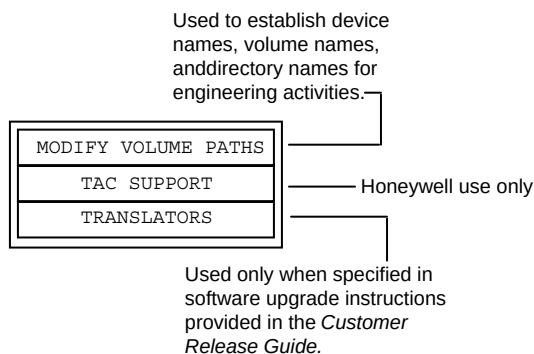
1.3 Command Processor Display and Support Utility Menu

Calling up these displays

When you select [COMMAND PROCESSOR] on the Engineering Main Menu, the Command Processor display appears. Use this display to enter and execute Command Processor commands.



When you select [SUPPORT UTILITES], the Support Utility Menu appears. This menu has the following targets:



For information about pathnames, see Pathnames, Volumes, Directories, and Files in section 1.5. For instructions for modifying volume pathnames, see:

- Modify Default Volume Pathnames Display (section 1.6),
- Modifying Default Volume Pathnames (section 2).

1.4 Uses for Command Processor Commands

What you can do with the commands

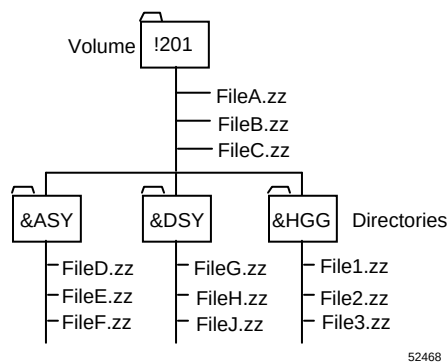
You can do any of the following:

- Establish volumes and directories on removable media
- Establish directories on HMs
- Copy volumes, files and directories from one medium to another. “Medium” means zip disk, cartridge disk, or emulated disk.
- Backup HM content on removable media and restore HM content from removable media
- Create, modify, and list file and directory descriptors
- Protect and unprotect files
- Display or print the content of files
- List attributes and status of files and devices
- Call up other Engineering Personality activities

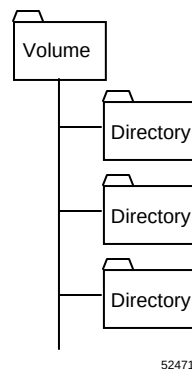
1.5 Pathnames, Volumes, Directories, and Files

Relationships

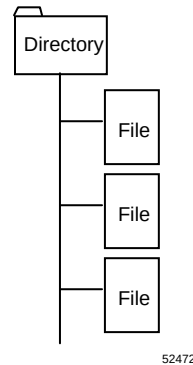
- A volume can contain its own files and one or more directories. If a volume does not have its own files, they are located in directories in the volume.
- Directories can contain one or more files. Files contain the data that you enter and use as you configure a system.



- Each removable media contains one volume, up to 63 directories, and no more than 9995 files. Zip cartridges, cartridge disks, and emulated disks are sometimes referred to as removable media.
- A History Module (HM) can have several volumes. Each volume can have up to 63 directories and up to 9995 files. HM volumes are established in Volume Configuration.
- A volume often contains several directories. One or more files may be associated with the volume itself.



- A directory usually contains a collection of related files. For example, one system directory (&ASY) contains the files related to network configuration. A common directory (CL) contains files related to user programs written in Control Language.



Pathname form

A pathname is simply a description of the location of a file. That description consists of these parts:

- Device Identity
- Volume Directory Name
- File Name
- LCN

Pathnames are often keyed in as part of a Command Processor command. Here is an example of a pathname:

```
$F2>HGPT>UNIT1A.DB
```

\$F2 is the Device Identity (Device ID). In this case, the device is the second Drive on this Console.

HGPT is the volume name or directory name.

UNIT1A.DB is the file name and suffix (extension).

DB is the suffix. It identifies the type of information in the file.

The Device ID can be in these additional forms:

```
PN:39>HPGT>UNIT1A.DB
```

```
NET>HGPT>UNIT1A.DB
```

PN:39 means, “physical device: node number 39.” It names the physical device whose LCN node number is 39.

NET means the file and volume or directory is in an on-line HM. This indicates that the LCN is the “device,” and whichever HM has the volume or directory is to respond.

LCN device path form includes the preceding and also these forms:

LCN ID \NET

LCN_ID \PN:nn

Pathname rules

It doesn't matter whether the letters in a pathname are capitals or lowercase letters; each is treated as a capital letter.

The Device ID is always in one of these forms: \$Fn, \$Pn, PN:nn, and NET.

\$Fn defines a removable media drive number.

\$Pn defines a printer device ID.

PN:nn defines a physical node number (LCN node number).

NET indicates that the volume or directory is to be found in an on-line HM on the LCN.

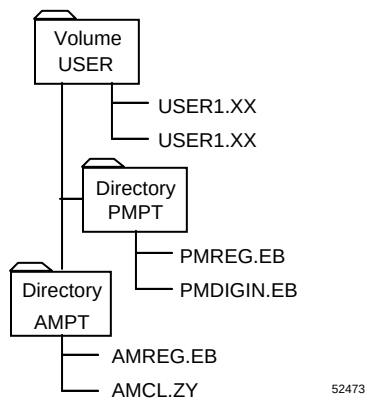
The Volume or Directory Name consists of One-to-four characters. “&” and “!” are reserved for volumes and directories provided by Honeywell.

While directories are subdivisions of volumes, pathnames can include either a volume name or a directory name in the volume/directory field. When trying to access a file, the pathname must contain the volume name or directory name where the file is located. In the following pathname:

NET>VDIR>FILE.XX

The VDIR field (one-to-four characters) must contain either the name of a volume in an HM on the network (LCN) or the name of a directory in a volume on the network.

If the file is in a directory, the system can find it through the directory name only, it does not need the volume name. In this example,



assuming volume USER is a volume on an HM, file USER2 . XY can be found through this pathname:

```
NET>USER>USER2 . XY
```

And file AMCL . ZY can be found through this pathname:

```
NET>AMPT>AMCL . ZY
```

But file AMCL . ZY cannot be found through this pathname:

```
NET>USER>AMCL . ZY
```

File names and suffixes

File names consist of one-to-eight characters. Suffixes (sometimes called “extensions”) are one or two alphanumeric characters, preceded by a period (dot). User-supplied suffixes begin with “X,” “Y,” or “Z.”

Suffixes serve to identify the type of information contained in the file. These are some of the more common system suffixes:

- AO=CL/AM object file
- BU=Backup file
- CF=NCF file
- CL=CL source file
- DA=Area database
- DB=Intermediate data file (IDF)
- DF=User DDB file
- DO=Schematic object file

- DS=Schematic source file
- DX=Print file from Picture Editor
- EB=Exception-build file
- EC=Execute Command file
- FO=Free-format log object file
- FS=Free-format log source file
- JL=Logic block listing file
- JO=Logic block object file
- JS=Logic block source file
- KO=Button configuration object file
- KS=Button configuration source file
- LE=CL error listing file
- LS=CL listing file
- PI=Personality image file
- PL=Parameter list file
- WA=Area working file

1.6 Modify Default Volume Pathnames Display

Default Volume Pathnames

The Engineering functions remember a set of Default Volume Pathnames. These pathnames indicate the location of the volumes and directories the Engineering functions use as you enter your configuration information.

Each removable media is a volume and has a volume name. HMs can have up to 30 volumes.

More information can be found in, *Engineer's Reference Manual*, Volumes and Types of Data Stored in HMs section.

The display

On the display that follows, you can see all of the major volume/directory names that the Engineering functions remember and use.

When you need to, you can change the pathnames on this display. The data-entry documents tell you when you need to do so. Instructions for changing the pathnames on this display are in section 2, Modifying Default Volume Pathnames.

Modify Default Volume Pathnames Display

DD:MM:YY HH:MM:SS 1

MODIFY DEFAULT VOLUME PATH NAMES

Edit all desired default paths and ENTER

HG GDF [NET>&HGG>]	NETWORK CONFIG [NET>&ASY>]	CL OVERLAY [NET>&OP2>]	DEF OVERLAY [NET>&OP1>]	SDT OVERLAY [NET>&OP4>]
HM/AM/CM GDF [NET>&AMG>]	CL SOURCE/OBJ [NET>CL>]	NOT USED [NET>&OP1>]	LBC OVERLAY [NET>&OP1>]	FIND NAMES OVLY [NET>&OP4>]
AREA DB GDF [NET>&ARG>]	CL PARAM LIST [NET>CL>]	NOT USED [NET>&OP1>]	TRANSLATORS OVLOAD NODE OVRLY [NET>&OP4>]	[NET>&OP4>]
CL CUSTOM GDF [NET>&CDSG>]	USER DEFLT PATH [NET>TEST>]	BUTTN CFG OVRLY [NET>&OP1>]	CONFIG OVRLY [NET>&OP1>]	PICTURE EDITOR [NET>&OP1>]
NIM GDF [NET>&NMG>]	KEY FILE VOLUME [NET>&KFO>]	SMCC OVERLAY [NET>&OP2>]	TAC SUPPORT OVIGENERIC OVRLAYS [NET>&OP5>]	[NET>&OVG>]
NIM GDF [NET>&NM2>]	EXT LOAD MODULE [NET>&CUS>]	DOC CTL DIR [NET>&DOC>]	TEMP FILE DIR [NET>TEIL>]	NCF BACKUP PATH [EX>&ASY>]
SET DEVICE PATH TO REM. MEDIA	SET DEVICE PATH TO "NET"	MAIN MENU	UTILITIES MENU	

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1.7 References

Document access

Throughout this document, references to other documents are made by title only.

- To access documents provided electronically, select the document title in the main topics list (select Main from the Topics button). Not all documents are available electronically. If the document is not listed in the Main Topics listing, please reference your paper documentation.
- To access anything in the paper documentation, find the title in the following document listing, and use the binder number provided there.

Document listing

Table 1-1 Document Listing

Document Title	Binder Number	Binder Name
System Startup Guide	3030 1	Implementation/Startup & Reconfiguration - 1
Data Entity Builder Manual	3032 1	Implementation/Engineering Operations - 1
Control Language/AM Data Entry	3035 2	Implementation/AM - 3
Engineer's Reference Manual	3030 2	Implementation/Startup & Reconfiguration - 2

2. Modifying Default Volume Pathnames

2.1 Overview

Section purpose

This section tells you how to modify the default volume pathnames that the Engineering functions use for most of its operations.

Section content

This section provides the following procedures:

- Preparing to Make Pathname Changes
- Choosing the Type of Pathname Change
- Changing All Pathnames to NET
- Changing All IDs to Removable Media IDs
- Changing Individual Pathnames

2.2 Preparing to Make Pathname Changes

Calling up the Modify Default Volume Pathnames Display

Start with the Engineering Main Menu.

If you are using the Engineering functions, return to the Main Menu (select the [MAIN MENU] target or hold <CTL> and press <HELP>).

If the Engineering functions are not running on this US, load the Universal Personality. Instructions are in the *System Startup Guide*.

Use the following procedure to call up the Modify Default Volume Pathnames display:

Table 2-1 Document Listing

Step	Action	Result
1	On the Engineering Main Menu, select [SUPPORT UTILITIES].	The Support Utility Menu appears.
2	Select [MODIFY VOLUME PATHS].	The Modify Default Volume Pathnames display appears.

Pathname example

The following is an example of one of the pathnames on this display:

Modify Default Volume Pathnames Display

DD:MM:YY HH:MM:SS 1

MODIFY DEFAULT VOLUME PATH NAMES

Edit all desired default paths and ENTER

HG GDF NET>&HGG>	NETWORK CONFIG NET>&ASY>	CL OVERLAY NET>&OP2>	DEB OVERLAY NET>&OP1>	SDT OVERLAY NET>&OP4>
HM/AMCM GDF NET>&AMG>	CL SOURCE/OBJ NET>CL>	NOT USED NET>&OP2>	LBC OVERLAY NET>&OP1>	FIND NAMES OVLY NET>&OP4>
AREA DB GDF NET>&ARG>	CL PARAM LIST NET>CL>	NOT USED NET>&OP2>	TRANSLATORS OVL NET>&OP4>	LOAD NODE OVRLY NET>&OP4>
CL CUSTOM GDF NET>&DSG>	USER DEFLT PATH NET>TEST>	BUTTN CFG OVRLY NET>&OP1>	CONFIG OVRLY NET>&OP1>	PICTURE EDITOR NET>&OP1>
NIM GDF NET>&NMG>	KEY FILE VOLUME NET>&KFO>	SMCC OVERLAY NET>&OP2>	TAC SUPPORT OVL NET>&OP5>	GENERIC OVRLAYS NET>&OVG>
NIM GBE NET>&NM2>	EXT LOAD MODULE NET>&CUS>	DOC CTL DIR NET>&DOC>	TEMP FILE DIR NET>TFIL>	NCF BACKUP PATH \$FX>&ASY>

SET DEVICE PATH TO REM. MEDIA SET DEVICE PATH TO "NET" MAIN MENU UTILITIES MENU

EXT LOAD MODULE
NET>&CUS

└─ Volume or Directory Name

└─ Device ID

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2.3 Choosing the Type of Pathname Change

Types of changes

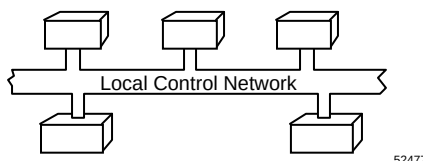
You may choose one of the following methods to change the Default Volume Pathnames:

- Changing All Pathnames to NET
- Changing All IDs to Removable Media IDs
- Changing Individual Pathnames

2.4 Changing All Pathnames to NET

How to make this change

To change all of the Default Volume Pathnames to specify HMs on the LCN, use the following procedure.



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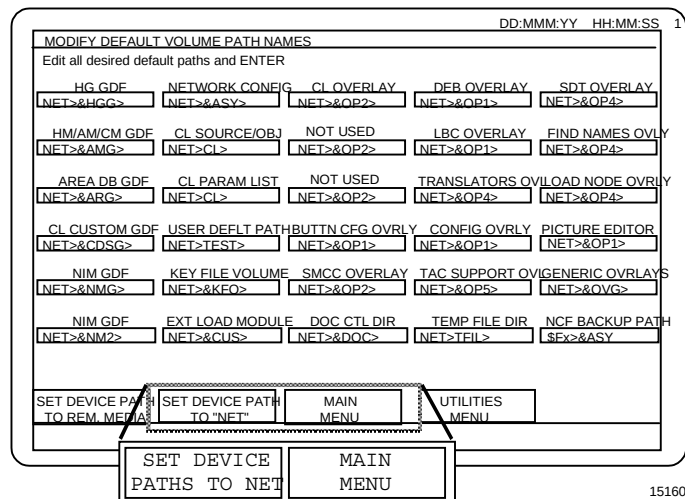
Table 2-2 Specify HMs on the LCN

Step	Action	Result
1	On the Modify Default Volume Pathnames display, select [SET DEVICE PATH TO "NET"] and press <ENTER>.	The device ID in all pathnames changes to NET in blue letters (except the NCF Backup Path for removable media).

Still need to change one or more individual NET pathnames?

Yes—see section 2.6, Changing Individual Pathnames.

No—select [MAIN MENU] to return to the Engineering Main Menu.



2.5 Changing All IDs to Removable Media IDs

How to make this change

To change all of the Default Volume Pathnames to removable media IDs (identification numbers), use the following procedure:

Table 2-3 Change to removable media device IDs

Step	Action	Result
1	On the Modify Default Volume Pathnames display, select [SET DEVICE PATHS TO REM. MEDIA], and press <ENTER>.	The device ID in all pathnames changes to \$Fm in blue letters.

Still need to change one or more individual pathnames?

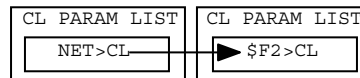
Yes—see section 2.6, Changing Individual Pathnames.

No—Select [MAIN MENU] to return to the Engineering Main Menu, shown above.

2.6 Changing Individual Pathnames

Making this change

To change an individual Default Volume Pathname,



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use the following procedure:

Table 2-4 Change an individual Default Volume Pathname

Step	Action	Result
1	Touch or tab to the pathname's port.	The cursor goes to that port.
2	Type over the device name, the volume/directory name, or both; or press <CLR-ENT> to clear the port, and key in a completely new pathname. Then press <ENTER>.	If there are no errors, the pathnames are redisplayed in blue.

3. Command Processor Commands

3.1 Overview

Section content

First, this section provides an alphabetical list of all of the Command Processor commands. You can use this list to look up each of the commands. This list is within section 3.2, Alphabetical List of Commands.

Next, this section tells you how to call up the Command Processor display and execute commands through that display. This is within section 3.3, Executing Commands.

Other sections of this document contain detailed descriptions of the commands with instructions for their use. Each of these sections contains commands in the following categories. They are as follows:

- File Commands, section 4
- History Module Commands, section 5
- Volume & Directory Commands, section 6
- Emulated Disk commands, section 7
- Print & Display Commands, section 8
- List & Status Commands, section 9
- Descriptor Commands, section 10
- Miscellaneous Commands, section 11
- Decision-Making Commands, section 13



ATTENTION

All keys mentioned in this document are keys on the Engineer's Keyboard.

3.2 Alphabetical List of Commands

Command Processor Commands

- Activate & Inactivate Point Commands, section 11.2
- Backup HM on Removable Medium, section 5.2
- Call Engineering Activities, section 11.3
- Copy File, section 4.2
- Copy Volume, section 6.2
- Create Directory, section 6.3
- Create Emulated Disk, section 7.2
- Data Entity Builder Execute Command (DEBEC), section 11.8
- Data Out, section 8.2
- Decision-Making Commands, section 13
- Delete Directory, section 6.5
- Delete Emulated Disk, section 7.7
- Delete File(s), section 4.3
- Dismount Emulated Disk, section 7.4
- Documentation Tool, section 11.4
- End Command Processor Session, section 11.5
- Equipment List Builder, section 11.6
- Execute Command, section 11.7
- Find Names, section 9.2
- Floppy (Removable Media) Copy, section 11.9
- Get US or HM Device Status, section 9.3
- List File Attributes (List & Status Commands), section 9.4
- List Volumes, section 9.5
- Load Multiple Nodes, section 11.10
- Lock Emulated Disk, section 7.5

- Modify File or Directory Descriptor, section 10.2
- Mount Emulated Disk, section 7.3
- Move File, section 4.4
- Pause, section 11.11
- Pre-Defined Documentation Tool Query, section 11.13
- Print or Display File Content, section 8.3
- Prompt Data, section 8.4
- Prompt Data Output, section 8.5
- Protect/Unprotect File, section 4.5
- Protect/Unprotect Zip Disk, section 11.12
- Rename File, section 4.6
- Restore HM from Removable Medium, section 5.3
- Schedule EC Execution (R530), section 11.14
- SCHEDULE Command, section 11.15
- Sequence Program Loading Command, section 11.16
- Set HM Disk Drive Off-Line, section 5.4
- Set User Pathname, section 11.17
- Set Volume Descriptor, section 10.3
- Short Cuts and “Wildcard” Characters, section 11.18
- Slew Printer Paper, section 8.6
- Synchronize HM Disk Drives, section 5.5
- Time Command, section 11.19
- Unlock Emulated Disk, section 7.6
- Unschedule Command, section 11.20
- Write Boot, section 11.21

3.3 Executing Commands

Calling up the Command Processor display

You can start with the Engineering Main Menu, or you can “escape” from one of the Engineering Personality activities to the Command Processor display.

If the Engineering Personality is not running on the Universal Station or Universal Station^X, this is either because the station is running the Operator Personality portion of the Universal Personality, or because the Engineering Personality is not loaded into the Universal Station or Universal Station^X.

If the Operator portion of the Universal Personality is running, call up the Engineering Main Menu by holding <CTL> and pressing <HELP>.

If the Engineering Personality is not loaded into the Universal Station or Universal Station^X, load it. For loading instructions, see the *System Startup Guide*.

If you are starting from the Engineering Main Menu, see Calling up the Command Processor from the Main Menu.

If you are in any Engineering Personality activity except Logic Blocks, see “Escaping” to the Command Processor.

Calling up the Command Processor from the Main Menu

With the Engineering Main Menu on the screen, select [COMMAND PROCESSOR]. The Command Processor display appears.

“Escaping” to the Command Processor

With a display for an Engineering Personality activity (except Logic Blocks) on the screen, press [ESC]. The Command Processor display appears.

When you are finished with the Command Processor, you can return to the display you “escaped” from by holding <CTL> and pressing <HELP>.

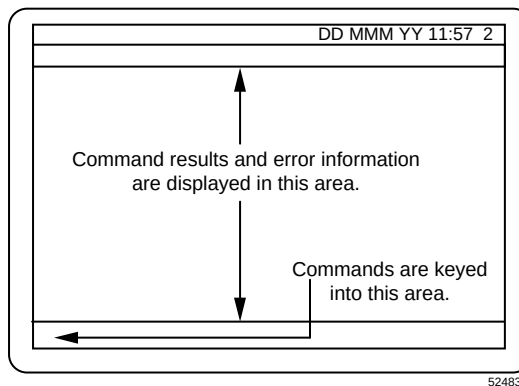


ATTENTION

All keys mentioned in this document are keys on the Engineer's Keyboard.

Executing a command

Areas on the Command Processor display are as follows:

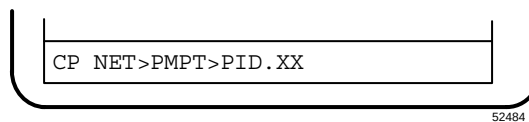


If you need information about the form of a command, you have two ways to find it.

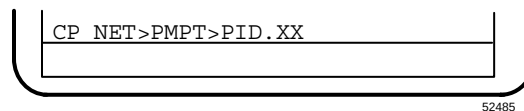
- Press [HELP]. The Command Processor help display appears. It shows the form of each command.
- Look up the command description in other sections of this document. You can use the 3.2 Alphabetic Lists of Commands to do this.

Entering a command

To enter a command, key the command into the entry area and press <ENTER>.



You will see an indication that the command is being executed. When execution is complete, a “complete” message appears.



If the command did not execute successfully, see 12.3 Recovering from Command Errors.



ATTENTION

Many of the commands take several seconds; in some cases, several minutes to complete. Usually, you can see that the command is executing. As a volume/directory is copied, each file is listed on the display as it is copied.

Reusing commands

To reuse a command line that you entered earlier, touch that line on the screen or use the <tab/cur> cursor movement keys to go to it and press [SELECT]. The command line is copied into the command area on the display. You can then press [ENTER] to execute the command, or make changes in the command area and then press [ENTER].

If the correct user volume already appears in the upper-right corner of the screen, you can omit the NET>VOL> portion of the pathnames.

4. File Commands

4.1 Overview

About the File commands

These commands copy, delete, transfer, move, protect, and rename files.

Commands in this category

The file commands are as follows:

4.2 Copy File

4.3 Delete File(s)

4.4 Move File

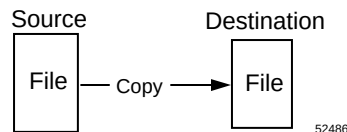
4.5 Protect/Unprotect File

4.6 Rename File

4.2 Copy File

Purpose

Copy a file and place the copy in a destination file. The destination file can be in the same volume or directory as the source file or in a different volume/directory. The source and destination files can be in volumes or directories in an HM or on a removable media.



If a file has a descriptor, when copied it follows the file. If a file descriptor is copied to a volume not supporting descriptors, it will be lost on the destination volume.

If a file not supporting file descriptors is copied to a volume that does, the destination file has a blank descriptor. For more information, see (Descriptor Commands) 10.2 Modify File or Directory Descriptor.

Command form

```
CP NET>VDIR>FILE.XX NET>VDIR>= -D
```

where

- = – Make the destination-file name and suffix equals to the source-file name and suffix.
- D– Indicate file copies and destination file.

Alternate command names are COPY, CPY, and COP.

The source-file name must include a suffix. The destination-file name cannot—the destination suffix is always the same as the source suffix.

Valid device ID forms are NET, &FN, PN: nn. The PN: nn form must be used if the volume/directory is on an HM that is running its Initialization Personality.

Message at completion

```
Copy Complete
```

Error messages

Volume Not found—The volume or directory you named for the source or the destination was not found where the device ID indicated it should be.

Insufficient Storage Space—There is not enough storage space left in the destination volume/directory. You may want to delete some files in that volume or directory to make enough space available.

Destination File Extension Not Allowed—The source file suffix (extension) and the destination file suffix must be identical.

Variations and options

CP NET>VDIR>FILE.XX \$F1>VDIR>FILE

where

NET>VDIR>FILE.XX– Source

\$F1>VDIR>FILE– Destination

CP NI\NET>VDIR>*.* NET>\$F1>VDIR>=

where

NI– By entering the Network Gateway ID, files can be copied from one LCN to another.

.– Copy all files with all suffixes from the source volume.

=– Make destination file names and suffixes equal to those on the source volume.

RULA variations and options

To specify a pathname to a device at a Remote RULA site, include the RULA Remote system ID notation (. . \) in the pathname.

For these examples, enter the command from either the configured Remote or Host station:

Copy the Host site NCF to the Remote site removable media:

CP NET>&ASY>NCF.CF . . \ \$f1>TEST>=

Copy a Picture Editor source file from the Remote site to the Host site:

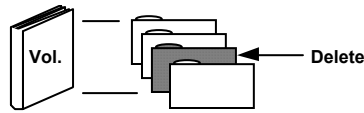
CP . . \ \$f1>TEST>RMPIC.DS NET>PICT>= -D

Reference: >>**RULA User Manual**

4.3 Delete File(s)

Purpose

Deletes one or more files from the specified volume. A deleted file and its data cannot be recovered.



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Command forms

```
DL NET>VDIR>FILE.XX
```

or

```
DL NET>VDIR>ORF.*
```

or

```
DL NET>VDIR>*. *
```

where

ORF.*– Delete all unprotected files named ORF, with any suffix.

.– Delete all files on this volume, with any suffix.

Alternate command names: DEL and DELETE

Valid device ID forms: NET, \$Fn, and PN:nn. The PN:nn form must be used if the volume/directory is on an HM that is running its Initialization Personality.

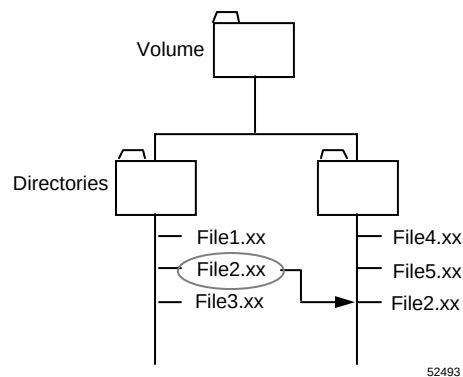
Message at completion

```
Delete complete
```

4.4 Move File

Purpose

Moves a file from one directory to another directory in the same volume. Also moves a file from a volume to a directory in the same volume, or from a directory to the volume that contains the directory.



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Command form

```
MV NET>DIR1>FILE.XX DIR2
```

or

```
MV $F1>DIR1>FILE.XX DIR2
```

or

```
MV $F1>DIR1>ALPHA.* DIR2
```

where

- | | |
|---------------------------|--|
| MV NET>DIR1>FILE.XX DIR2 | – Move FILE.XX from DIR1 to DIR2 |
| MV \$F1>DIR1>FILE.XX DIR2 | |
| MV \$F1>DIR1>ALPHA.* DIR2 | – Move all the unprotected files named ALPHA, with any suffix. |

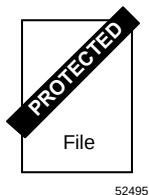
Message at completion

Move Complete

4.5 Protect/Unprotect File

Purpose

Protects a file or files from being overwritten, deleted, or moved. Also cancels such protection.



Command form

```
PT NET>VDIR>FILE.XX
```

where

FILE.XX – Protect this file

```
PT NET>VDIR>HIC.*
```

where

HIC.* – Protect all HIC files, all suffixes

```
PT or UNPT NI\NET>VDIR>USER.*
```

where

NI\NET>VDIR>USER.– Protect or unprotect all
* USER files, including all
suffixes on another LCN as
indicated by the Network
Gateway ID

```
UNPT NET>VDIR>HIC.*
```

where

HIC.*– Cancel protection for all HIC files, all
suffixes

Valid device ID forms are NET, \$Fn, PN:nn. The PN:nn form must be used if the volume is on an HM that is running its Initialization Personality.

Alternate command names are PROT, PRO, UNPROT, and UNPRO.

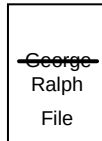
Message at completion

File Protect Complete or Unprotect Complete

4.6 Rename File

Purpose

Gives a new name to a file or a volume. The original name is no longer recognized by the system, unless it is given to some other file or volume.



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Command form

where

GEORGE.* RALPH – Change the name of
GEORGE files, all suffixes,
to RALPH. Files names can
be up to eight characters.

RN NET>VDIR>GEORGE.* RALPH

To rename a removable media volume:

where

VOL> LOV> – Change the name of
volume VOL to LOV.
Volume and directory names
can be up to four characters.

RN \$FN>VOL> LOV>

An alternate form of the command name is REN.

Valid device ID forms are NET, \$Fn, and PN:nn. The PN:nn form must be used if the volume/directory is on an HM that is running its Initialization Personality.

Message at completion

Rename Complete

5. History Module Commands

5.1 Overview

Commands in this category

The History Module commands are as follows:

5.2 Backup HM on Removable Medium

5.3 Restore HM from Removable Medium

5.4 Set HM Disk Drive Off-Line

5.5 Synchronize HM Disk Drives



ATTENTION

You can use the [Get US or HM Device Status command \(section 9.3\)](#), to list the status of History Module disk drives.

5.2 Backup HM on Removable Medium

Purpose

Copies all the directories and files on an HM onto a removable media. Continuous-history volumes, journal volumes, and the System Maintenance Journal are not copied.

During the backup operation, a volume can be automatically created on the removable medium. Appropriate directories are also created. If the backup requires more than one removable medium, a prompt requests that a new one be mounted. If needed, a sequential volume name can be created on the new medium (for example, &B01, &B02, &B03, and so on). Each new medium must be mounted in the same drive.

The backup data stored by this command is suitable for placing back on the HM with the Restore Command.

Reference: 5.3 Restore HM from Removable Media.

Using the Backup command



ATTENTION

This command creates an Execute Command file on the destination medium, which you can modify to create a custom backup operation that backs up only the volumes you specify. The .EC file contains one or more Copy Volume commands that do the actual backup. The file name is in this form: BKUPnn .EC, where nn is the HM node number.

References:

Section 11.7, Execute Command (Miscellaneous Commands)

Section 6.2, Copy Volume (Volume & Directory Commands)

We recommend that you use just-initialized cartridges or floppies as the destination media for this command.

Use the Create Volume command to initialize the media. We suggest you use -MF 3000 for cartridges and -MF 300 for floppies, and we suggest -BS 300 for either type of medium.

Reference: (Volume & Directory Commands)

Write down the volume name you used with the Create Volume command. You will need that name if you use the EC command to execute the .EC file on the destination cartridge.

Reference: Section 11.7, Execute Command (Miscellaneous Commands)



CAUTION

Before you start a backup

Ensure that no HM operation takes place that could alter the content of any of the files to be copied during the backup operation. Automatic checkpointing from all checkpointed nodes (HG, NIM, CG, and AM) must be disabled and no demand checkpoints allowed.

If you must format a removable medium during the backup operation, remember that any information stored on that media would be lost.

Command form

```
BACKUP PN:43 $F1
```

This example creates a file BKUP43.EC on the removable media.

Alternate command names: BACK, BKUP .

Response of the Backup command

The .EC file is created first, then it is executed (if not inhibited as indicated within “Variations and Options.”

This command file consists of one Copy Volume command for each volume to be copied from the HM; therefore, the Backup command responds just as the Copy Volume command does. It is the Copy Volume command that requests new media to be mounted, as needed. The Copy Volume command also formats (or reformats) each medium, if needed, according to your responses to its prompts.

Reference: Section 6.2, Copy Volume (Volume & Directory Commands)

Message at completion

Backup complete

Variations and options

BACKUP PN:43 \$F1 -EC \$F2>BKUP>BACKUP43.EC

where

- | | |
|-----------------------|--|
| -EC | – Create the .EC file, only do not execute it. |
| \$F2>BKUP>BACKUP43.EC | – Pathname that defines where the .EC file is to be stored. |
| -NO_FMD | – No file descriptors are copied (unless -EC is used; the resulting EC file that is executed depends on the destination volume). |

User-modification of Backup .EC files

You can modify the .EC file created by the Backup command to create a backup operation that backs up only the volumes you specify. To do so, use the Text Editor to delete the CPV command lines you don't want.

This is an example of the .EC file:

&P

CPV PN:43>!001> &1 -A -D

CPV PN:43>!101> &1 -A -D

Where:

-D equals the destination device option

This is an example of the .EC command line:

```
EC NET>BKUP>HM1.EC $F1
```

Where:

\$F1 equals the destination device ID

For information about parameters in .EC files, see 11.7 Execute Command.



ATTENTION

When you use the Execute Command to execute the .EC file, you must use the volume name used by the Backup command when it generated the file.

Format a Zip Disk or 150MB Bernoulli Cartridge

To format a Zip disk or 150 MB Bernoulli cartridge to attempt to fit all the files in a HM during a Backup Command, do the following:

1. LS PN:nn>*>*. * where nn = node number. Let this run until finished and notice how many files are in the HM.
2. Use this number of files, plus about 100 extra and rounded out, to format the cartridge with the -X option, for example:

```
CR $F1>VOL> -F _MF 10000 -BS 1000 -X
```

5.3 Restore HM from Removable Medium

Purpose

Restores the contents of a volume with all of its directories and files from a removable media to an HM. The volume and directory must already exist on the HM. It will exist on the original HM, if the HM has not been initialized and has not had a failure that destroyed the volume or directory since the HM was prepared as in Tasks 12 through 16 in the *System Startup Guide*.



ATTENTION

This command creates an .EC file on the source medium. The .EC file has one Copy Volume command that copies the volume on the source medium to the HM.

References:

Section 11.7, Execute Command

Section 6.2, Copy Volume

You can modify the .EC file to facilitate restoring a volume to an HM where the HM volume is so large that two or more cartridges or floppies are needed to back it up. With such a modified .EC file, the volume can be easily restored with one EC command. The .EC file name is in this form: RESTVOLM.EC

Command form

```
RESTORE $Fn PN:nn
```

where

\$Fn— Source drive

nn— HM node number. Do not use node pair number.

Response to the Restore command

The .EC file is created first, then it is executed (if not inhibited). This command file consists of a Create Volume command that restores the content of one removable media; therefore, the Restore command responds just as the Copy Volume command does.

Reference: Section 6.4, Create Volume (Volume & Directory Commands)

If two or more cartridges or floppies are storing the backup information, you should modify the command file (.EC) by adding a Copy Volume command line for each subsequent medium, and placing PAUSE statements between CPV command lines. Each PAUSE statement should be preceded by a message to the screen that asks for the next removable media to be mounted.

The following is an example of a modified .EC file, the user adding the third, fourth, and fifth lines:

&P

CPV &1>&ASY> NET -D

&P Mount next medium and press ENTER

PAUSE

CPV &1>&ASY> NET -D

where

&P– This command sends a message to the screen, in this case, “Mount next medium and press ENTER.”

PAUSE– This command line waits for the user to mount the next medium and press ENTER.

CPV– This command line copies additional HM volume &ASY information from the next removable media, which in this example has the same name (&ASY) as the first. Additional HM volume information can be restored by repeating this command line sequence.

&1>– This is a variable removable-media parameter referencing the first parameter on the .EC command line (example shown below) that contains the \$Fn drive number:

EC NET>ECVL>RESTBB01.EC \$F1

Message at completion

Restore Complete

5.4 Set HM Disk Drive Off-Line

Purpose

Changes the status of the specified HM disk drive to off-line. This command is not valid unless the status for the specified disk drive is OK and it has a redundant partner that is OK.

This command can also be used to abort a disk synchronization operation.

Reference: Section 5.5, Synchronize HM Disk Drives

Command form

OFF PN: nn n

where

nn— HM node number

n— Number of HM drive to be set off-line

Indication of completion

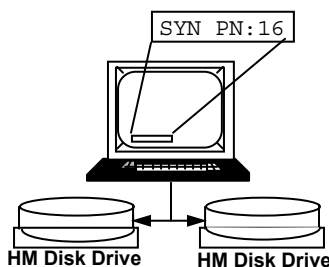
Use a List & Status Command (section 9.3) to verify that the status for the specified drive changed to OFFLINE.

To regain disk redundancy, execute a 5.5 Synchronize HM Disk Drives command.

5.5 Synchronize HM Disk Drives

Purpose

Synchronizes the data on both drives of a redundant pair of drives on a History Module. Synchronization takes place while the HM is running its on-line (&HMO) personality. Once synchronized, the same data is maintained on each drive by recording the same information on both of them at each write operation.



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Command form

SYN PN:nn

where

nn— HM node number

Operation

Synchronization is accomplished by copying of the most current good volume to each drive of the redundant pair. Normally one of the two drives contains the source for all volume data; however, in Release 400 and later, it is possible with the improved History Module Redundancy function, to cross synchronize drives.



ATTENTION

We strongly recommend that you synchronize HM disk drives only according to the following conditions:

- Both drives have been running on line before the execution of the SYN command.
- If the secondary drive has been off-line, but power has not been removed.
- If one drive fails or must be replaced for any reason. The replacement must be a blank, formatted disk drive. [Use the Hardware Verification Test System (HVTs) to format the disk drive.]

A redundant partner drive (secondary) that has had power removed while some activity is done to it on-line (volumes added or deleted), must be formatted before it can be synchronized.



ATTENTION

Because in Release 400 and later the HM disk drives are redundant on a volume-by-volume basis, it is important to realize that the History Module may be fully functional while there are corrupted volumes on any or all of the disks; therefore, before setting any disks off-line for replacement, you should reallocate bad sectors on the good drive [using the System Maintenance Control Center (SMCC) in the Engineering Personality] and then synchronize the disks (if possible) to ensure the best possible data integrity.

Cross synchronization means that some source data is taken from the primary and copied to the secondary and some source data is taken from the secondary and copied to the primary.

Synchronization can take several minutes to more than hour. It depends on how busy the HM is. The US is available for other functions as soon as synchronization is initiated.

To monitor the status of the HM during the synchronization, use the List & Status Commands (section 9.3). While the synchronization is in process, the status is SYN_IP and upon completion it becomes OK.

To abort the synchronization, use the Set HM Disk Drive Off-Line command (section 5.4) to change the drive status to OFFLINE.

Indication that synchronization is complete

When synchronization is complete, the HM's status, on the Node Status display, changes from SEVERE to OK and a message is sent to the Real Time Journal.

6. Volume & Directory Commands

6.1 Commands in this Category

Overview

The volume & directory commands are as follows:

Command form

```
CPV NET>VOLM $F2>VOLM -A -D -I
```

where

NET>VOLM— Source

\$F2>VOLM— Destination

-A— Include all directories in the volume
if the destination is a removable
media..

-D— List each file as it is copied.

Valid device ID forms are NET, \$Fn, PN:nn. The PN:nn form must be used if the volume is on an HM that is running its initialization Personality.

Message at completion

Copy Volume Complete

- Copy Volume, section 6.2
- Create Directory, section 6.3
- Create Volume, section 6.4
- Delete Directory, section 6.5

6.2 Copy Volume

Purpose

Copy all of the directories and files in a volume and place the copies in a destination volume. The destination volume can have the same name as the source volume or it can have a different name. The source and destination volumes can be in an HM, on a cartridge, or on removable media. Copy volume does not create directories on the HM.



ATTENTION

Recommendation: Use just-initialized cartridges or floppies as the destination media for this command. Use the Create Volume command to initialize the media. We suggest you use -MF 3000 for cartridges and -MF 300 for floppies, and we suggest -BS 300 for either type of medium.

Reference: Section 6.4, Create Volume

If you are making backup copies of personality media (Univ, Opr, Eng, Maint), after you copy the first media, you must execute a Write Boot command to that media.

Reference: Section 11.21, Write Boot (Miscellaneous Commands)

Command form

```
CPV NET>VOLM $F2>VOLM -A -D -I
```

where

NET>VOLM– Source

\$F2>VOLM– Destination

–A– Include all directories in the volume if the destination is removable media.

–D– List each file as it is copied.

Valid device ID forms are NET, \$Fn, PN:nn. The PN:nn form must be used if the volume is on an HM that is running its initialization Personality.

Message at completion

```
Copy Volume Complete
```

Variations and options

If you included the -I option on the command line and the medium you mounted is not formatted or does not have the expected volume name, this message appears:

```
Do you want to FORMAT/INIT this media? (VOLM)
```

where

(VOLM) - Press ENTER if this volume name is acceptable, key in a different volume name, or key in “NO,” to bypass the formatting function.

If you choose to format (initialize) the medium, this message appears:

```
Do you really want to FORMAT/INIT this media?
```

Press <ENTER> to continue the formatting operation and the Copy Volume operation, or key in “NO” to abort the formatting operation. If you press <ENTER>, any data in the destination volume will be overwritten.

The formatting operation is as described in section 6.4, Create Volume.

During the volume copying process, this message may appear:

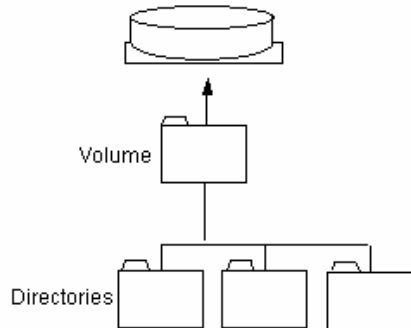
```
Please mount next removable media on $Fn.
```

To continue the operation, mount another removable media in the same drive.

6.3 Create Directory

Purpose

Creates a directory under a volume, A directory is a subdivision of the volume. Each directory can have one or more files assigned to it.



Each removable media is a volume. An HM can have several volumes. A volume can have up to 63 directories.

It is not necessary to use this command before using the Backup HM on Removable Medium command.



ATTENTION

If you need to create the same directories on a volume from time-to-time, for example, directories that will store demand checksums, consider creating an Execute Command File with the necessary Create Directory command lines in it and then executing an Execute Command to execute them all at once.

Command form

The form is:

```
CD $F1>VOL DIR
```

```
CD NET>VOL DIR
```

A space is required between the pathname and directory name. A directory name can be up to four characters long.

Message at completion

```
Create Directory Complete.
```

6.4 Create Volume

Purpose

Initializes removable media, which deletes any existing volumes, directories, and files, and gives the removable media a new (or the same) volume name (up to four characters). Includes a Format option that marks and numbers the sectors on the media so that the system can read from and write to the media.

It is not necessary to use the Format option to initialize (create) media that has already been formatted. When you do not use the format option, the media remains formatted as before.



ATTENTION

HM volumes are established in Volume Configuration after the disks(s) have been initialized.



CAUTION

With or without the Format option, the Create Volume command destroys any data stored on the removable media.

Command form

```
CR $F1>VOL -F -MF 3000 -BS 1700 -X
```



ATTENTION

–MD and –FD options do not work with –X.

where:

- F (-FMT) – Format option for media
- MF 3000 – Maximum Files option (can be as high as 32767, but keep it reasonable to just 100 more than what you need)
- BS 1700 – Block Size option for transfer size. 1700 is maximum without File Descriptors (-FD) option. With –FD the maximum for –BS is 600.
- MD – Memory-Resident Directory option
- FD – File Descriptors option
- X – Extended Number of Directories beyond the usual 63 allowed, up to 2046 directores (for LCNs with TPN R510.7 and later releases).

Alternate command name forms are CRE, CV, and CREATE.



ATTENTION

- The `-X` option can only be used on removable media.
- **Do not** use the `-X` option to create removable media that will be used for the following purposes:
 - Checkpointing
 - Translating (When using the removable media for translations, such as during software upgrades).
 - When also using either the Memory-Resident Directory (`-MD`) and/or the `-FD` options.
- You cannot use the following commands on removable media that has been created with the `-X` option:
 - Find Name (FN) command on checkpoints
 - AM/CL read/write extensions (no file I/O or CDS read/write allowed to removable media with `-X`).

Message at completion

Create Volume Complete

Error messages

Create Volume error messages are as follows:

- Max Files Out of Range (1-9995)—This message appears if a value of 0 is entered. The `-MF` option calls for a number in the range from 1 to 32767; the message appears if the number of files exceeds 32767.
- Block Size Out of Range—The `-BS` option called for a number not in the range from 26 to 1700 if file descriptors are not being used. If file descriptors are being used, the `-BS` option called for a number not in the range of 26 to 600.

Variations and Options

```
CR $F1>VOLM> -F -MF 300 -MD -FD -BS 26
```

where

- F- Format Option
- MF 300- Maximum Files Option
- MD- Memory-Resident Directory Option
- FD- File Descriptor Option
- BS 26- Block Size Option



ATTENTION

The -MD option is not used with the -BS option. If you try this, the following message appears: -BS OPTION NOT NEEDED WITH -MD.

Typical use

```
CV $F1>VOL1  
or  
CR $F1>VOL1
```

Option descriptions

- **Format option**—The form is **-F**. The Format option marks and numbers the sectors of the disk so the system can read from and write to it. This option was initially designed for floppy disks. For removable media, this option is no longer needed.
- **Max Files option**—The form is **-MF nnn**, where nnn is a number from 1 to 32767. “nnn” specifies the maximum number of files that can be stored on removable media. The more files, the longer it takes to search the media directory. If you know the number of files you need, use the -MF option to reduce the number of files, allowing faster transfers. The default value for the maximum number of files on removable media (cartridge or zip) is 3020.



CAUTION

You cannot increase the maximum number of files, if you should need them later, without destroying all of the data on the removable media. It is better to specify a number somewhat greater than your estimated needs.

- **Block Size option**—The form is **-BS nnn**, where nnn is a number from 26 to 1700. This option specifies the maximum amount of directory information that can be transferred at one time (in one block). The larger the block size, the faster the access. However, because read/write operations are executed in the node, more node memory is used. The default value equals the **-MF** value or 300, whichever is less. (The **-BS** value does not limit the number of files that can be transferred, only the amount of directory information in one internal transfer operation. For example, if the block size is 300 and 300 files are present, only one transfer is needed; if the block size is 30, ten transfers are needed.) With file descriptor, the limit is 600. See File Descriptor option.
- **File Descriptor option**—Form is **-FD**. This reserves space for the file descriptors for all files and directories on this volume. If you do not use this option with Create Volume, you will not be able to copy descriptors to the media. For more information, see 10.2 Modify File or Directory Descriptor. Do not use this option with the **-X** Extended Directory Option because it will cancel the option.
- **Memory-Resident Directory option**—Form is **-MD nnn**. This option stores the entire media directory in the node memory in one block, allowing slightly faster transfers. It is similar to using a block size equal to the number of files. This option automatically sets the block size. Do not use this option with the **-X** Extended Directory Option because it will cancel the option.
- **Extended Number of Directories option** – To get beyond 63 directories (in one Volume) in a removable media disk or Bernoulli, use **-X** with R510.7 or later releases. This allows 2046 directories, which may be enough, when using the **-MF** option correctly, to allow most or all of a single HM drive to backed up to one Zip disk or 150 MB Bernoulli cartridge. Do not also use **-MD** and/or **-FD** options when using the **-X** option, since it will ignore the **-X** option.



CAUTION

The Extended Number of Directories option takes significant room in memory. Proceed with caution before using it.

Format a Removable Media

To format a removable media to attempt to fit all the files in a HM during a Backup Command, do the following:

1. **LS PN:nn>*>*. *** where nn = node number. Let this run until finished and notice how many files are in the HM.
2. Use this number of files, plus about 100 extra and rounded out, to format the cartridge with the **-X** option, for example:

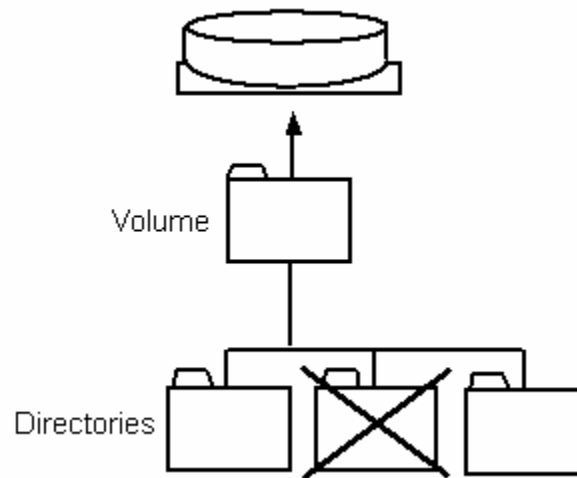
CR \$F1>VOL -F -MF 10000 -BS 1000 -X

6.5 Delete Directory

Purpose

Deletes a directory from a volume. A directory that still contains files cannot be deleted. If you have no further use for the directory's files, use Delete File command to delete them.

Reference: Section 4.3, Delete File(s) (File Commands)



Command form

```
DD NET>VOL DIR
```

Message at completion

```
Delete Directory Complete
```


7. Emulated Disk Commands

7.1 Overview

About Emulated Disk commands

With R670, emulated disks can be operated from the US/GUS Native Window using the Command Processor and EC files.

With R671, emulated disks can also be operated from Experion Station-TPS.

See *System Components Configuration Utility User's Guide, TP20W* for more information about emulated disk operations.

Configuration prerequisites

The configuration prerequisites for operating emulated disks using Command Processor and EC files are as follows:

Source node	Target node
US/GUS R670 or later	GUS R360 and Native Window loaded with GUS UxS personality R670 or later.



ATTENTION

The source and the target nodes must be on the same console.

Emulated disk path

On the target GUS node, a specific path is used for creating, mounting, and deleting emulated disks. This path is known as the emulated disk path and it can be configured using the Set Path button on the Emulated Disks page in Native Window. The default emulated disk path is:

C:\Program Files\Honeywell\TPS\Emulators\EmulatedDisks

For information about configuring the emulated disk path, see *System Components Configuration Utility User's Guide, tp20w*.

Commands in this category

The emulated disk commands are as follows:

7.2 Create Emulated Disk

7.3 Mount Emulated Disk

7.4 Dismount Emulated Disk

7.5 Lock Drive

7.6 Unlock Drive

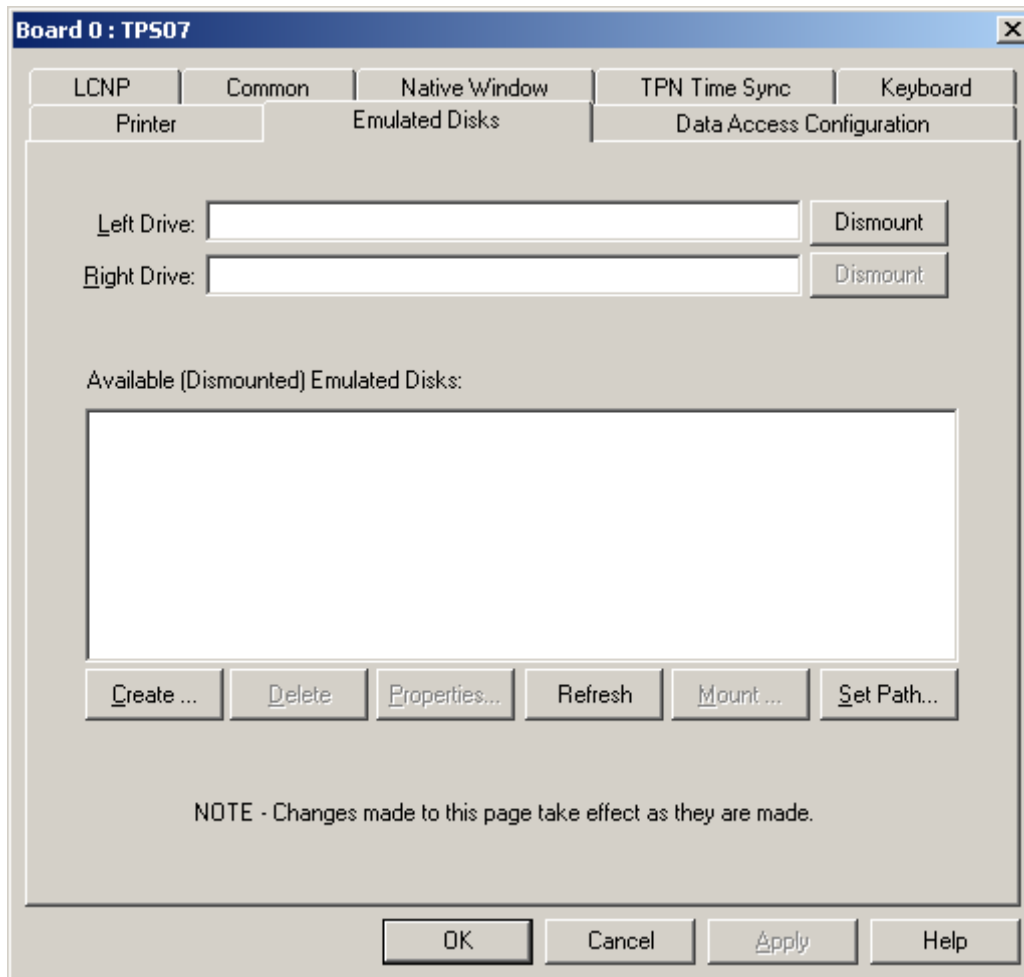
7.7 Delete

7.2 Create Emulated Disk

Purpose

The EDCREATE (Emulated Disk Create) command is used for creating an emulated disk on a local or a remote GUS node. The file is created in the emulated disk path. Optionally, you can also specify the size of the emulated disk and the allocation type.

Before creating an emulated disk, ensure the emulated disk path is configured in the target GUS node. This path can be configured using the Set Path button on the Emulated Disks page.



For information about configuring the emulated disk path, see *System Components Configuration Utility User's Guide, tp20w*.

The emulated disk created using the EDCREATE command is granted read and write permissions. In addition, the emulated disk appears in the "Available Emulated Disks" list on the Emulated Disks page.

Command form

EDCREATE SN:n <filename> [-FS <size>] [-D | -R]

Where

- n = Configured Station number of the target node within the console.
- You can refer to Console Status Display for the Station number.
- filename = Name of the emulated disk to be created.
- FS <size> = Size of the emulated disk to be created.
- Range = 1 to 500 MB.
- Default size = 20 MB.
- [-D | -R] = Allocation type.
- D indicates the allocation type is dynamic. That is, allocation varies depending on the actual size of the file.
- R indicates the allocation type is reserved.
- The default allocation type is reserved.

Alternate command forms include – EDCR, EDC.

Variations and options

EDCREATE SN:7 EMVL1

Where

- 7 = Configured Station number of the target node within the console.
- EMVL1 = Name of the emulated disk.

By default, 20 MB is allocated for the emulated disk.

EDCREATE SN:10 EMVL2 -FS 200

Where

- 10 = Configured Station number of the target node within the console.
- EMVL2 = Name of the emulated disk to be created.
- FS 200 = Size of the emulated disk. The emulated disk is allocated 200 MB space.

EDCREATE SN:2 EMVL3 -FS 100 -D

Where

- 2 = Configured Station number of the target node within the console.
- EMVL3 = Name of the emulated disk to be created on the target GUS node.
- FS 100 = Size of the emulated disk is 100 MB.
- D = Dynamic.

This indicates the space required for the emulated disk is dynamically allocated. However, the maximum space that can be allocated to the emulated disk cannot exceed 100 MB.

EDCREATE SN:8 EMVL4 -FS 300 -R

Where

- | | | |
|---------|---|--|
| 8 | = | Configured Station number of the target node within the console. |
| EMVL4 | = | Name of the emulated disk to be created on the target GUS node. |
| -FS 300 | = | Size of the emulated disk. |
| -R | = | Reserved. |

This indicates that 300 MB is reserved for the emulated disk. This space is allocated to emulated disk irrespective of the actual space occupied by the emulated disk.

Message at completion

Create Emulated Disk Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDCREATE command in EC files.

EDCREATE SN:6 EMVL3 -Q2

Error messages

The EDCREATE error messages are as follows:

Extra characters on line – The command contain extra characters which are not part of the syntax.

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Command too long – The total length of the command including the emulated disk path exceeds 140 characters.

Error creating LCN file – The hard disk space on the GUS node is not enough to create the emulated disk.

Incomplete command – The name of the emulated disk or any other necessary argument is missing.

Invalid allocation type – Allocation type is specified with both options -D and -R.

Invalid command format – The command arguments (SN:n or FS) are not in the correct form.

Invalid emulated disk path – This error may occur because of one of the following:

- Hard disk space on the target GUS node is not sufficient to create the emulated disk.
- The emulated disk path does not exist on the target GUS node.
- The emulated disk path has access restrictions.

Invalid file size – The file size specified is not in the range of 1 to 500 MB.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

Invalid LCN filename – The name of the emulated disk is misspelled or extra characters are included in the filename.

Invalid station number – The target Station is not configured in the console.

LCN file already exists – The emulated disk already exists on the target GUS node.

Node is not alive – The Station is not operational.

7.3 Mount Emulated Disk

Purpose

The EDMOUNT (Emulated Disk Mount) command is used for mounting an emulated disk on a specific drive in a GUS node. Before mounting an emulated disk, you must ensure the following:

- The emulated disk is already created in the emulated disk path.
- The drive on which the emulated disk must be mounted is empty. That is, ensure the drive is not locked and does not have any other emulated disk mounted.

On the Emulated Disks page, the emulated disk is removed from the list of “Available Emulated Disks” list. In addition, the emulated drive cannot be used for mounting any other emulated disk.

Command form

EDMOUNT <drive> <filename>

Where

Drive = Left/right drive of the target GUS node. This must be specified as “\$Fn” where “n” is a configured drive on the console.

You can refer to Console Status Display for the drive number.

filename = Name of the emulated disk to be mounted on the drive.

Alternate command forms include – EDMNT, MNT.

Variations and options

EDMOUNT \$F1 EMVL1

Where

\$F1 = A configured drive on the console.

EMVL1 = Name of the emulated disk to be mounted.

Message at completion

Mount Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDMOUNT command in EC files.

```
EDMOUNT $F1 EMVL1 -Q3
```

Error messages

The EDMOUNT error messages are as follows:

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Command too long – The total length of the command including the emulated disk path exceeds 140 characters.

Drive already mounted – The drive is locked or another emulated disk is already mounted on the drive.

Extra characters on line – The command contain extra characters which are not part of the syntax.

File already mounted – The emulated disk is already mounted on another drive.

Incomplete command – The name of the emulated disk or any other necessary argument is missing.

Invalid command format – The first argument – “\$Fn” is not in the correct form.

Invalid drive number – The drive is not configured on the console.

Invalid emulated disk path – The emulated disk path does not exist or the path has access restrictions.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

LCN file not found – The emulated disk is not found in the emulated disk path of the target GUS node.

7.4 Dismount Emulated Disk

Purpose

The EDDISMOUNT (Emulated Disk Dismount) command is used for dismounting an emulated disk from a specific drive in a GUS node. Before dismounting an emulated disk, you must ensure the drive is not locked.

On the Emulated Disks page, the emulated disk reappears in the “Available Emulated Disks” after dismounting. In addition, the drive is made available for mounting.

Command form

EDDISMOUNT <drive>

Where

Drive = The drive from which an emulated disk must be dismounted. This must be specified as “\$Fn” where “n” is a configured drive on the console.

You can refer to Console Status Display for the drive number.

Alternate command forms include – EDDMNT, DMT.

Variations and options

EDDISMOUNT \$F1

Where

\$F1 = The drive from which the emulated disk must be dismounted.

Message at completion

Dismount Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDDISMOUNT command in EC files.

EDDISMOUNT \$F1 -Q2

Error messages

The EDDISMOUNT error messages are as follows:

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Extra characters on line – The command contain extra characters which are not part of the syntax.

Incomplete command – Command arguments are missing.

Invalid command format – The first argument – “\$Fn” is not in the correct form.

Invalid drive number – The drive is not configured on the console.

Invalid Emulated Disk path – The emulated disk cannot be dismounted because the emulated disk path has access restrictions.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

Drive locked – The emulated disk cannot be dismounted because the drive is locked.

Drive already dismounted – The emulated disk is already dismounted.

7.5 Lock Drive

Purpose

The EDLOCK (Emulated Disk Lock) command is used for locking a drive on which an emulated disk is mounted. Locking prevents dismounting of an emulated disk from a specific drive. The drive can be unlocked using the EDUNLOCK command. In addition, the drive is automatically unlocked if the target GUS node is restarted. However, the drive remains locked when Native Window is reloaded.

On the Emulated Disks page, a locked drive is disabled for dismounting.

Command form

EDLOCK <drive>

Where

Drive = Left/right drive of a GUS node that must be locked.
This must be specified as “\$Fn” where “n” is a configured drive in the console.

You can refer to Console Status Display for the drive number.

Alternate command forms include – EDLCK, EDLK.

Variations and Options

EDLOCK \$F1

Where

\$F1 = The drive to be locked.

Message at completion

Lock Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDLOCK command in EC files.

EDLOCK \$F1 -Q3

Error messages

The EDLOCK error messages are as follows:

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Drive is empty – The drive is empty. It does not have any emulated disks mounted.

Drive already locked – The drive is already locked.

Extra characters on line – The command contain extra characters which are not part of the syntax.

Incomplete command – Command arguments are missing.

Insufficient privileges for operation – The drive cannot be locked because of insufficient access privileges.

Invalid command format – The first argument – “\$Fn” is not in the correct form.

Invalid drive number – The drive is not configured on the console.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

7.6 Unlock Drive

Purpose

The EDUNLOCK (Emulated Disk Unlock) command is used for unlocking a locked drive. Locked drives are automatically unlocked if the target GUS node is restarted.

On the Emulated Disks page, the unlocked drive is enabled for dismounting. Note that emulated drives cannot be locked or unlocked from the Emulated Disks page. If a drive is locked, you must unlock it using the EDUNLOCK command.



ATTENTION

If emulated drives are locked in the GUS node using EDLK command, you cannot unlock them (using command prompt or EC file) by loading that node with the Universal personality.

R670 requires the target node to be a GUS node (Native Window loaded from 'W'). Otherwise, the Emulated Disk commands targeted on that node returns an "Invalid functionality on node" error.

Command form

EDUNLOCK <drive>

Where

Drive = Left/right drive of a GUS node that must be unlocked. This must be specified as "\$Fn" where "n" is a configured drive on the console.

You can refer to Console Status Display for the drive number.

Alternate command forms include – EDUNLK, EDUL.

Variations and options

EDUNLOCK \$F1

Where

\$F1 = The drive to be unlocked.

Message at completion

Unlock Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDUNLOCK command in EC files.

```
EDMOUNT $F2 -Q2
```

Error messages

The EDUNLOCK error messages are as follows:

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Extra characters on line – The command contain extra characters which are not part of the syntax.

Incomplete command – Command arguments are missing.

Insufficient privileges for operation – The drive cannot be locked because of insufficient access privileges.

Invalid command format – The first argument – “\$Fn” is not in the correct form.

Invalid drive number – The drive is not configured on the console.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

Drive already unlocked – The drive is already unlocked.

7.7 Delete Emulated Disk

Purpose

The EDDELETE (Emulated Disk Unlock) command is used for deleting an emulated disk on the target GUS node. Before deleting the emulated disk, you must ensure the emulated disk is already dismounted. In addition, the emulated disk must be available in the emulated disk path.

Command form

EDDELETE SN:n <filename> [-N]

Where

- | | | |
|----------|---|---|
| nn | = | Configured Station number of the target node within the console. |
| | | You can refer to Console Status Display for the Station number. |
| filename | = | Name of the emulated disk to be deleted. |
| -N | = | Option to remove the emulated disk only from “Available Emulated Disks” list. |

Alternate command forms include – EDDEL, EDDL.



CAUTION

If the [-N] option is not used, the EDDELETE command permanently deletes the emulated disk. Hence, evaluate your requirements and then delete the emulated disk.

Message at completion

Delete Emulated Disk Complete

Variation for EC files

In EC files, use the -Qn option to determine the success of the operation. The value of “n” must be in the range of 1 to 32. The following is an example of using the EDDELETE command in EC files.

EDDELETE SN:1 -Q32

Variations and options

EDDELETE SN:10 EMVL1

Where

10 = Configured Station number of the target node within the console.

EMVL1 = The emulated disk to be deleted.

EDDELETE SN:9 EMVL2 -N

Where

9 = Configured Station number of the target node within the console.

EMVL2 = The emulated disk to be deleted.

-N = Option to remove EMVL2 only from “Available Emulated Disks” list.

Error messages

The EDDELETE error messages are as follows:

Command timed out – The command is not carried out because there was no response from the target node within 10 seconds.

Command too long – The total length of the command including the emulated disk path exceeds 140 characters.

Error deleting LCN file – The emulated disk is already mounted on another drive.

Extra characters on line – The command contain extra characters which are not part of the syntax.

Incomplete command – The command arguments are missing.

Invalid command format – The first argument – “SN:n” is not in the correct form.

Invalid Emulated disk path – The emulated disk path does not exist or the path has access restrictions.

Invalid functionality on node – The target GUS node is a pre-R360 node or it does not have the UxS personality.

Invalid station number – The target Station is not configured in the console.

LCN file not found – The emulated disk is not found in the emulated disk path of the target GUS node.

Node is not alive – The Station is not operational.

7.8 Using Emulated Disk Commands for Backup

Example

Consider a scenario where you want to backup the volume – NET>VOLM on an emulated disk on a remote GUS node. The follow details are available:

- Station number of the target GUS node = 8
- Empty drives – \$F5 and \$F6
- Space required for the backup – 500 MB

To backup NET>VOLM, use the following sequence of commands in EC files:

```
EDCREATE SN:8 EMVL1 -FS 500 -D -Q10
&IF ^Q10 &THEN EDMOUNT $F5 EMVL1 -Q11
&IF ^Q11 &THEN CR $F5>BCK -FMT -Q12
&IF ^Q12 &THEN CPV NET>VOLM $F5>BCK -D -Q13
&IF ^Q13 &THEN EDDISMOUNT $F5
```

A similar sequence of commands can be used in an EC file, which can be scheduled as required.

To restore the backup, use the following command:

```
RESTORE $F5 NET>BCK
```


8. Print & Display Commands

8.1 Overview

About the Print & Display commands

These commands direct the outputs of other commands to the Universal Station or Universal Station^X screen or to a printer, request the printing or display of text file content, and slew the paper in a selected printer one page at a time.

Commands in this category

The print and display commands are as follows:

- Data Out, section 8.2
- Print or Display File Content, section 8.3
- Prompt Data, section 8.4
- Prompt Data Output, section 8.5
- Slew Printer Paper, section 8.6

8.2 Data Out

Purpose

Specifies which device(s) to send data to when subsequent Print or List commands are executed. The data always goes to the Universal Station screen, and it can be directed to a printer or to a file in a volume/directory in an HM, or on removable media.

Command form

DO—Send data to the screen, only. This cancels the effect of the preceding Data Out commands.

DO \$P2—Send the data to the screen and to printer 2.

DO NET>HGPT>ORF.xx—Send the data to the screen and to HM file ORF.xx in volume/directory HGPT.

Message at completion

Data Out Complete.

Error messages

Device Error—You keyed in a device ID that was incorrect or named a device that doesn't exist or isn't operating.

8.3 Print or Display File Content

Purpose

Displays the content of the specified file on the Universal Station and, if the last Data Out command so specified, prints that same information on a printer.

Command form

P NET>VDIR>FILE.XX

Valid device ID forms are NET, \$Fn, PN:nn. The PN:nn form must be used if the volume/directory is on an HM that is running its Initialization Personality.

Alternate command names: PR, PRINT

Message at completion

Print Complete

Variations and options

P NET>VDIR>ORF.*

where

ORF.*– Display (and print) the content of all
ORF files, all suffixes.

P NET>VDIR>*.XY

where

*.XY– Display (and print) the content of all files
on this volume/directory with a suffix of XY.

8.4 Prompt Data

Purpose

Lets you direct the prompters and error messages generated by the Command Processor to both the Universal Station screen and a printer, or only to the screen.

If you specify Prompt Data On and if the last Data Out command specified a printer, the prompters and error messages are included in the printouts. If you specify Prompt Data Off, the prompters and error messages won't be printed, regardless of the last Data Out command.

Command form

PD ON—Allow prompters and error messages to be printed.

PD OFF—Do not allow prompters and error messages to be printed.

Message at completion

Prompt Data Turned On (Off)

8.5 Prompt Data Output

Purpose

Sends command lines, completion messages, and error messages to a file or to a printer. Also can cancel such outputs.

Command form

PO \$P1—Send command lines, completion messages, and error messages to printer 1.

PO NET>VDIR>FILE.XX—Send command lines, completion messages, and error messages to this file.

PO—Cancel Prompt Data Output.

Valid Device ID forms are NET, \$Fn, PN:nn. The PN:nn form must be used if the volume is on an HM that is running its Initialization Personality.

Message at completion

Promptout Complete

8.6 Slew Printer Paper

Purpose

On the printer specified by the last Data Out command, advances the paper by one page.

Command form

SLW

Message at completion

Slew Complete

9. List & Status Commands

9.1 Overview

About the List & Status commands

These commands generate printed or displayed lists of system data and status information.

Commands in this category

The List & Status commands are as follows:

- Find Names, section 9.2
- Get US or HM Device Status, section 9.3
- List File Attributes, section 9.4
- List Volumes, section 9.5

9.2 Find Names

Purpose

Searches for objects that are part of other objects. This command is used to display or print lists of items, such as entity names or CDS names that are contained in objects such as checkpoint files, history groups, CL blocks, and user files. For example, you could use Find Names to obtain a list of all of the parameters in entity A100 that point to other entities or to a list of all of the entities that CL block VLRATIO refers to.

The last Data Out command determines if the output is displayed only, or is displayed and printed.



ATTENTION

Instructions for use of Find Names are provided in the *Find Names* document.

Command form

FN object identifier, object identifier, name identifier
where

name identifier– Optional



ATTENTION

You can either key in a Find Names command line or use the Find Names displays to automatically generate the command line. Instructions for use of Find Names are provided in the *Find Names* document.

Message at completion

Find Names Complete

Error messages

If the requested Find Names search cannot be completed, a message appears to explain why. Here are two examples:

No hiway checkpoint was found in the specified volume:

NO MATCH FOR HWY_CP *

“ENTITY” is misspelled. The “^” points to the invalid object:

9.3 Get US or HM Device Status

Purpose

Provides a list of the devices connected to this Universal Station and the status of some internal software modules, if no node number is specified. If the HM node number is specified, this command provides the status of the HM disk drives.

Command form

STA—List the status of the devices attached to each US in this console.

STA PN:nn—List the status of the disk drive(s) in the History Module whose node number is nn.

STA NET—List the status of the disk drives in all HMs on the LCN.

STA PN:nn -D

or

STA NET -D

where

-D = Include the virtual volume listing for the HMs listed.

An alternate form of the command name is STATUS.

Message at completion

Status List Complete

Status list examples

US status list created by entering STA.

LDID	NODE	VDID
\$F1	01	FD00
\$C01FD01	01	FD00
\$F2	01	FD01
\$C01FD02	01	FD01
\$P1	01	LP00
\$C01LP01	01	LP00
\$T1	01	I000
\$NCF_DATA	00	LP00
\$LOAD_DATA	00	I000
\$PRINTER	02	LP00
\$TRENDPENS	02	LP00
\$RTJ	02	
\$C01LP01	02	

Status List Complete

Status list for an HM with one disk drive created by entering STA NET.

Node	Peripheral	Device Address	Redundant Device	Status	System Name
16	FIXED_DISK	5	NONE	OK	NONE

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Status list for an HM with four drives created by entering STA NET -D.

Node	Peripheral	Device Address	Redundant Device	Status	System Name
43	Fixed Disk	2	3	OK	NONE
43	Fixed Disk	3	2	OK	NONE
43	Fixed Disk	4	5	OK	NONE
43	Fixed Disk	5	4	OFFLINE	NONE
VVID	STATUS 5	STATUS 4	VVID	STATUS 3	STATUS 2
!001	OK	OK	TEST	OK	OFFLINE
!901	OK	OK	USER	OK	OFFLINE

Virtual volume information listed when you include the -D option.

Drive 5 is offline.

Drives 4 and 5 and drives 2 and 3 are redundant.

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9.4 List File Attributes

Purpose

Lists the attributes of a file, or files, associated with the specified volume or directory. The list appears on the US screen and on any printer, or in any file specified by the last Data Out command.

Reference: Section 8.2, Data Out

Command forms

LS NET>VOL—List the attributes of all files in directory VOL.

LS NET>DIR>FILE.LS—List the attributes of the file whose name is FILE.LS.

LS \$F1>DIR—List the attributes of all files in directory DIR.

LS NET>*>* .XX List the attributes of all files in all volumes and directories on the LCN that have the suffix “XX.”

LS NET>A*>* .XX List the attributes of all "XX" files in directories whose names start with "A."

Alternate command names are LIST and CAT.

Valid Device ID forms are NET, \$Fn, and PN:nn. The PN:nn form must be used to list files in an HM running its initialization personality (&HMI).

Message at completion

Catalog Complete

Variations and options

LS NET>VOL -D

where

-D— Do not list files, list only the volume,
directory, sectors, and files information.

LS NET>VOL>FILE.*

where

*— List the attributes of each FILE file, all suffixes.

LS NI\NET>VOL

where

NI– List the attributes of files located on another LCN by entering the Network Gateway ID.

LS \$F1>> or LS \$F1

where

LS \$F1>> or LS \$F1– List the attributes of all files in the removable media volume in drive 1 (do not know the volume name).

LS NET>VOL -A

where

-A– List all files associated with this volume and all files associated with the directories in this volume. Output is truncated at 132 characters if it exceeds this length.



ATTENTION

If the listing lines are too wide to fit on the screen, you can shift the display left or right by holding <CTL> and pressing <L> or <R>.

LS NET>VOL -FD

where

-FD – List all files associated with this volume or directory along with their file descriptors.

LS NET>VOL -REC -FD

where

-REC– List all files associated with this volume or directory, along with the START and END fields.

-REC -FD– List all files associated with this volume or directory, along with their record information and file descriptors. This option combination deletes the START and END fields so that 132 characters of information can be displayed on one line.

LS NET>VOL -N

where

-N– List all associated with this volume or directory, but only display file information. Header (title) or footer (number of sectors or files used) information is not included in the output. You can also type out the -N as NO HEADING, NOHDG, or NOHEADING to identify the option in printed reports.

LS NET>VOL -D

where

-D– Header (title) or footer (number of sectors or files used) information is included in the output. The Detail option can be used only with the -FD and -ALL options or by itself. You can also type out the -D as DETAIL to identify the option in printed reports.

LS NET>USER -S DIR, NAME, EXT, DATE, DESC

where

- S – Sorts files attribute output by either directory, file name, extension, date, or descriptor. The option default is to sort by the directory name. More than one sort option can be used. You can also type out the -S as SORT to identify the option in printed reports.

LS NET>* -M "PATTERN_MATCH_SPECIFIER"

where

- M – Displays only those files whose descriptors match what is entered between the quotes. Wild card characters of * and ? are allowed. Sets [] can also be used where group of characters are listed and one character is used (for example, a[bcd]e matches abe, ace, or ade). You can also type out the -M as MATCH to identify the option in printed reports.

LS NET -P DATE, TIME

where

- P – Lists those files whose date and/or time is prior to what is entered. You can also type out the -P as PRIOR to identify the option in printed reports. Wildcard characters are not allowed.

Date=MO/DAY/YR

Time=HR:MN

LS NET -F DATE, TIME

where

- F – List those files whose date and/or time is at or after what is entered. You can also type out the -F as FROM to identify the option in printed reports.

LS NET -BF

where

- BF – Displays limited attribute information.
You can also type out the -BF as BRIEF
to identify the option in printed reports.

Rules

The output listings include device and directory fields. The Network Gateway identifier is not included in the output unless it is specified in the command line.

Permission to access files through the Network Gateway by using the LS command is configured in NCF Configuration.

Wildcard rules are as follows:

- Wildcard characters “*” and “?” can be included in directory names. With or without wildcard characters, directory names cannot exceed four characters. For example,
 - LS NET >*>*.DB specifies listing of all volumes and files with the .DB suffix on the LCN network.
 - LS NET>A*>*.DB specifies listing of all .DB files in all directories whose names start with “A.”
 - LS NET>M??N>*. * specifies listing of all files in directories whose names start with “M” and end in “N.”
- The pattern specified with the -M(ATCH) option can include wildcard characters “*,” “?,” and “[“ with “]” The brackets define sets of characters. For example,
 - a [bcd] e would match “abe,” “ace,” or “ade.”

Legal combinations of options are as follows:

- The -D(ETAIL) option can be combined only with -FD and -A(LL).
- If both -F(ROM) and -P(RIOR) are used, the dates must be different.
- The -B(RIEF) option cannot be combined with -P(RIOR), -F(ROM), or -S(ORT).

Listing examples

**ATTENTION**

To see how to interpret the lists produced by the LS command, see [Interpreting the LS Listings](#).

Single file.

```
LS NET>CL>VLR.LS
NET>CL>VLR.LS
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP      #RECS  RECSIZ
-----
NET      CL  VLR      LS    L          5  11/19/85  14:59      71  -----

FILES LISTED: 1

Catalog Complete                                     52642
```

Multiple files.

```
LS NET>TEST>*. *
NET>TEST>*. *
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP      #RECS  RECSIZ  #BLKS  BI
-----
NET  TEST  HG001HG1  CP   C    0   03/08/90  13:12  ----  -----  1424  --
NET  TEST  HG001HG2  CP   C    1   03/12/90  09:47  ----  -----  1424  --
NET  TEST  HG001MAS  CP   C    2   03/12/90  09:47  ----  -----    1  --
NET  TEST  HG001051  CP   C    2   03/08/90  13:12  ----  -----   10  --
NET  TEST  HG00105L  CP   C    1   03/12/90  09:47  ----  -----    4  --
NET  TEST  HG001052  CP   C    4   03/12/90  09:47  ----  -----   10  --

FILES LISTED: 6

Catalog Complete                                     52643
```

Wildcard characters specifying all files in all volumes and directories.

```
LS NET>*>*. *
NET>*>*. *
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP      #RECS  RECSIZ  #BLKS  BLKS]
-----
NET    &001  001FILE1  XX   . . . . .
NET    &001  001FILE2  XX   . . . . .
NET    &001  001FILE3  XX   . . . . .
NET    &ASY  ASYFILE1  YY   . . . . .
NET    &ASY  ASYFILE2  YY   . . . . .
NET    &ASY  ASYFILE3  YY   . . . . .
NET    &DSY  DSYFILE1  XX   . . . . .
NET    &DSY  DSYFILE2  YY   . . . . .
NET    &DSY  DSYFILE3  ZZ   . . . . .

FILES LISTED: 9

Catalog Complete                                     52644
```

File descriptor.

```

LS NET>*.* -FD
NET>*.* -FD
DEVICE DIR  FILENAME  EXT TYP P VER TIME STAMP          FILE DESCR:
-----
NET    &001 001FILE1  XX  . . . . .
NET    &001 001FILE2  XX  . . . . .
NET    &001 001FILE3  XX  . . . . .
NET    &ASY ASYFILE4  YY  . . . . .
NET    &ASY ASYFILE5  YY  . . . . .
NET    &ASY ASYFILE6  YY  . . . . .
NET    &DSY DSYFILE1  XX  . . . . .
NET    &DSY DSYFILE2  YY  . . . . .
NET    &DSY DSYFILE3  ZZ  . . . . .

FILES LISTED: 9

Catalog Complete                                     52645

```

Record information and file descriptors.

```

LS NET>ASY>*.* -REC -FD
NET>&ASY>*.* -REC -FD
DEVICE DIR  FILENAME  EXT TYP P VER TIME STAMP #RECS RECSIZ #BLKS
-----
NET    &ASY ASYFILE1  XX  . . . . .
NET    &ASY ASYFILE2  XX  . . . . .
NET    &ASY ASYFILE3  XX  . . . . .
NET    &ASY ASYFILE4  YY  . . . . .
NET    &ASY ASYFILE5  YY  . . . . .
NET    &ASY ASYFILE6  YY  . . . . .
NET    &ASY ASYFILE7  XX  . . . . .
NET    &ASY ASYFILE8  YY  . . . . .
NET    &ASY ASYFILE9  ZZ  . . . . .

FILES LISTED: 9

Catalog Complete                                     52646

```

No title option.

```

LS NET>TEST>*.* -N
NET>TEST>*.* -N

NET    TEST TSTFILE1  XX  L  0 03/04/91 12:01 53 -----
NET    TEST TSTFILE2  XX  C  0 05/21/91 10:16 14 -----
NET    TEST TSTFILE3  XX  C  5 06/18/91 09:05 28 -----
NET    TEST TSTFILE4  YY  C  2 04/10/91 11:22 25 -----
NET    TEST TSTFILE5  YY  L  3 04/11/91 01:04 36 -----
NET    TEST TSTFILE6  YY  L  4 07/16/91 08:35 49 -----
NET    TEST TSTFILE7  XX  C  3 02/04/91 03:41 62 -----
NET    TEST TSTFILE8  YY  C  1 09/30/91 07:06 19 -----

FILES LISTED: 8

Catalog Complete                                     52647

```

Sort option.

```

LS NET>USER>*.* -S EXT
NET>USER>*.* -S EXT
DEVICE DIR  FILENAME  EXT TYP P VER TIME STAMP          #RECS RECSIZ #BLKS I
-----
NET    USER FILE1111  XX  L  0 4  03/04/91 12:01 53 ----- 1
NET    USER FILE2222  YY  C  4 5  04/10/91 11:22 25 ----- 5
NET    USER FILE3333  ZZ  L  1 0 10/13/91 12:32 33 ----- 1

FILES LISTED: 3

Catalog Complete                                     52648

```

Sort option, two fields sorted.

```

LS NET>USER>*. * -S EXT DATE
NET>USER>*. * -S EXT DATE
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP          #RECS  RE
-----
NET  USER  FILE1111  XX   L   0   4   03/04/91 12:01    53   --
NET  USER  FILE1298  XX   L   2   4   05/28/91 10:06   ---  --
NET  USER  FILE2222  YY   C   4   5   04/10/91 11:22    25   --
NET  USER  FILE1901  YY   C   3   4   09/21/91 09:42   179  --
NET  USER  FILE4904  ZZ   C   1   5   08/16/91 08:12   ---  --
NET  USER  FILE3333  ZZ   L   1   0   10/13/91 12:32    33   --

FILES LISTED: 6

Catalog Complete                                     52649

```

Match option.

```

LS NET>USER>*. * - M "ALPHA"
NET>USER>*. * - M "ALPHA"
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP          FILE DESCF
-----
NET  USER  FILE5555  XX   L   0   0   03/04/91 12:01  ALPHA
NET  USER  FILE7777  XX   C   4   2   04/10/91 11:22  ALPHA
NET  USER  FILE8888  YY   L   1   1   10/13/91 12:32  ALPHA

FILES LISTED: 3

Catalog Complete                                     52650

```

Prior option.

```

LS TEST>*. * -P 04/20/91
NET>TEST>*. * -P 04/20/91
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP          #RECS  REC
-----
NET  TEST  TESTF2   XX   L   0   1   03/11/91 12:01    53   --
NET  TEST  TESTF6   YY   C   4   0   04/13/91 11:22    25   --
NET  TEST  TESTF10  ZZ   L   1   2   04/19/91 12:32    33   --

FILES LISTED: 3

Catalog Complete                                     52651

```

From date option.

```

LS TEST>*. * -P 04/20/91
NET>TEST>*. * -P 04/20/91
DEVICE DIR  FILENAME  EXT  TYP  P  VER  TIME  STAMP          #RECS  REC
-----
NET  TEST  TESTF2   XX   L   0   1   03/11/91 12:01    53   --
NET  TEST  TESTF6   YY   C   4   0   04/13/91 11:22    25   --
NET  TEST  TESTF10  ZZ   L   1   2   04/19/91 12:32    33   --

FILES LISTED: 3

Catalog Complete                                     52652

```

Brief listing option.

```

LS NET>TEST>*. * -B
NET>TEST>*. * -B
DEVICE DIR  FILENAME  EXT  TYP
-----
NET      TEST  TSTFILE1  XX   .   .
NET      TEST  TSTFILE2  XX   .   .
NET      TEST  TSTFILE3  XX   .   .
NET      TEST  TSTFILE4  YY   .   .
NET      TEST  TSTFILE5  YY   .   .
NET      TEST  TSTFILE6  YY   .   .

FILES LISTED: 6

Catalog Complete

```

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Interpreting the LS listings

The following two figures show how to estimate file size. A sector and a block are both 256 bytes.

Estimating file size when BLKSIZ is not shown with a contiguous file.

FILE NAME	EXT	TYP	P	#	RECS	RECSIZ	#BLKS	BLKSIZ	START	END	TIME STAMP
HG001HG1	CP	C					1212		54	1265	04/01/89 14:23
HG0010501	CP	C	*				10		1271	1280	03/26/89 14:42
HG00105L	CP	C					4		1266	1269	03/26/89 14:42
HG001HG2	CP	C					1212		1282	2493	03/26/89 14:42
HG001MAS	CP	C					1		1270	1270	03/26/89 14:43

Protected files indicated by "*" in the "P" column.

VOLUME: &701
 VOLUME DESC: HG Chkpt Control
 MAX FILES: 321
 FILES LISTED: 05
 TOTAL SECTORS: 7680
 SECTORS IN USE: 2493
 ACCESS LEVEL: NETWORK READ WRITE
 CURRENT DIRECTORY ASSIGNMENT:
 &Z38 &Z39

Maximum number of files in this volume

Total sectors allocated for this volume

Number of sectors currently in use—Usually somewhat more than END - START, e.g., 2493 - 54 = 2439, which is less than 2493.

HG001HG2.CP uses (2493-1282)+1 = 1212 sectors * 256 = 310,272 bytes.

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Estimating file size when BLKSIZ is shown with a linked file.

INDEX.AG uses (4*13)+1 = 14 sectors, 223*14 = 237.

FILE NAME	EXT	TYP	P	#	RECS	RECSIZ	#BLKS	BLKSIZ	START	END	TIME STAMP
CA0000SY	CM	C					215		8	222	03/20/89 15:10
INDEX	AG	L			1	3106	1	13	223	223	03/20/89 15:57
AGG	AG	L			6	100	1	09	237	237	03/20/89 15:57
SECOND	AG	L			1	1940	1	08	247	247	03/20/89 15:57
PAG	PA	L	*		1	104	3	04	256	256	03/20/89 15:57
ERR_BUFF	ER	L			1536	04	4	04	279	279	03/20/89 15:58

Protected files indicated by "*" in the "P" column.

VOLUME: !401
 VOLUME DESC: Maintenance Support
 MAX FILES: 26
 FILES LISTED: 06
 TOTAL SECTORS: 1248
 SECTORS IN USE: 306
 ACCESS LEVEL: NETWORK READ WRITE
 CURRENT DIRECTORY ASSIGNMENT:
 !ESY

For Honeywell internal use only.

PAG.PA uses 3*4 = 12 sectors, 256 * 12 = 3072 bytes.

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9.5 List Volumes

Purpose

Lists the volumes and directories in the HM named in the command line or in all HMs and, if the last Data Out command (section 8.2) so specified, prints the list on a printer. In release R400 and later, LV NET shows the volumes and directories for the History Modules running either their operating personality (&HMO) or their initialization personality (&HMI).

Command form

LV PN:nn

where

nn– List volumes in this HM node number.

LV NET

where

NET– List volumes in all HMs on this LCN.

An alternate command name is LSV.

Message at completion

List Volumes Complete

Volume list example

```
HM:43
!001 ( !A01 !A02 !A03 )
&101 ( &B01 &HG0 &BO3 &AMO &B04 &OPR &ENG &OV1 &OV2 & OV3 &OV4 )
&001 ( &ASY &DSY &HGG &AMG &ARG )
└── Volumes ───────────────────┘ └── Directories ───────────────────┘
```

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10. Descriptor Commands

10.1 Overview

About the Descriptor commands

These commands allow you to add and to edit descriptors that are attached to volumes, directories, and files. These descriptors appear in the printed and displayed listings generated by other commands, such as List File Attributes (LS) and List Volumes (LV).

Commands in this category

The Descriptor commands are as follows:

- Modify File or Directory Descriptor, section 10.2
- Set Volume Descriptor, section 10.3

10.2 Modify File or Directory Descriptor

Purpose

Establishes, changes, or deletes file or directory descriptors on volumes that have file descriptors enabled. Any removable media may have descriptors enabled. They are enabled by using the -FD option of the Create Volume command. System volumes, Area Database volumes, and user volumes on an HM may have file descriptors enabled during volume configuration.

Descriptors remain with files and directories when they are copied. If they are copied to a volume that does not support them, they are lost on the destination volume.

Descriptors are blank if a file or directory that does not support them is copied to a volume that does support them.

You may find it helpful to have descriptors attached to files or directories that you list with the List File Attributes command or the List Volumes command. If a whole list item with its descriptor extends beyond the width of the screen, you can shift the display to the left or right by holding <CTL> and pressing <L> or <R>.

References:

Section 9.4, List File Attributes (List & Status Commands)

Section 9.5, List Volumes (List & Status Commands)

Command forms

- To add or modify a file descriptor: MFD NET>VDIR>FILEA.XX FILE DESCRIPTOR STRING.
- To add or modify a directory descriptor: MFD NET>VDIR DIR DESCRIPTOR STRING.
- To delete a file or directory descriptor, execute a Modify Descriptor command without a descriptor string.



ATTENTION

Volume descriptions are established or modified by the Set Volume Descriptor command.
Reference: [Section 10.3, Set Volume Descriptor](#)

Message at completion

Modify File Mgr Desc Complete

10.3 Set Volume Descriptor

Purpose

Establishes or changes the volume descriptor for the specified volume. The volume descriptor appears in the detailed form of the lists created by the List File Attributes command. It can be any combination of up to 20 printable characters. You can use it to remind you of the purposes of your user volumes.

```
.EB FILES, BOILER_B20A
```

Spaces and underbars are
allowed as word separators.

52660

Command form

SVD NET>VDIR>Descriptor String

Message at completion

Set Volume Descriptor Complete

Error messages

Incomplete Command—The volume descriptor was left out of the command line.

Extra Characters On Line—More than 20 characters were keyed into the volume descriptor field of the command line, with an embedded space at position 21 and additional characters at position 22 and beyond.

Volm Desc Must Be <= 20 Chars—More than 20 characters with no spaces were keyed into the volume descriptor field of the command line.

11. Miscellaneous Commands

11.1 Overview

About the Miscellaneous commands

The commands in this category are those that do not belong to one of the other command categories.

Commands in this category

- The miscellaneous commands are as follows:
- Activate & Inactivate Point Commands, section 11.2
- Call Engineering Activities, section 11.3
- Documentation Tool, section 11.4
- End Command Processor Session, section 11.5
- Equipment List Builder, section 11.6
- Execute Command, section 11.7
- Data Entity Builder Execute Command, section 11.8
- Floppy (Removable Media) Copy, section 11.9
- Load Multiple Nodes, section 11.10
- Pause, section 11.11
- Protect/Unprotect Zip Disk, section 11.12
- Pre-Defined Documentation Tool Query, section 11.13
- Schedule EC Execution (R530), section 11.14
- Schedule Command, section 11.15
- Set User Pathname, section 11.17
- Short Cuts and “Wildcard” Characters, section 11.18
- Time Command, section 11.19
- Unschedule Command, section 11.20
- Write Boot, section 11.21

11.2 Activate & Inactivate Point Commands

Description

ACTIVATE and INACTIVATE, available for R430 and later, are Data Entity Builder commands that can activate and inactivate any point in the system that has an execution state. These commands are usually used in EC (Execute Command) files. But they also can be issued from the Command Processor.

Definitions

ACTIVATE point command allows the user to activate a point that is currently in the inactive state. INACTIVATE point command allows the user to inactivate a point this is currently in the active state.

Form of commands

These two commands use the following form variations when they are used in EC files or typed into the command line of the Command Processor:

Table 11-1 Point Commands

ACTIVATE	INACTIVATE
ACTIVATE + point name [-c]	INACTIVATE + point name [-c]
ACTIVE + point name [-c]	INACTIVE + point name [-c]
ACTIV + point name [-c]	
ACT + point name [-c]	INACT + point name [-c]
Then press the Enter key.	Then press the Enter key.

where

-c – the confirmation message option.

When this option is used with the ACTIVATE and INACTIVATE commands, the Command Processor provides a confirmation message that the point has been made active or inactive.

Example of EC file

These commands can be entered separately on the command line, or can be used in an EC file, intermixed with other EC commands. A sample EC file with ACTIVATE and INACTIVATE commands appears on the next page:

```
INACT A100 -c
UNLK CLPROG1 A100
```

```
CLK CLPROG1 A100
ACTIV A100 -c
INACT B100
UNLK CLPROG2 B100
CLK CLPROG2 B100
ACTIV B100
```

Confirmation messages will appear, as shown below, as the EC finishes processing these commands.

```
A100 SET INACTIVE
A100 SET ACTIVE
```

Error messages

Error messages are displayed for these commands when the point specified:

- Does not exist. (“Invalid point ID”)
- Does not have a point execution status (PTEXECST) parameter (“Cannot Activate/Inactivate Pt.”)

Change state journaling

No journaling occurs when an engineer changes a point’s execution state using these commands issued from the Command Processor or an EC file.

Journaling does occur when an operator changes the point execution status (Active/Inactive targets) on the Point Detail Display.

11.3 Call Engineering Activities

Purpose

Twenty-three subcommands are available to call up Engineering activities (overlays) from the Command Processor display. You can use these subcommands to go directly to one of the activities without returning to the Engineering Main Menu.

Each of the subcommands calls for the appropriate software overlay. Such overlays can be called only if you call the Command Processor display from the Engineering Main Menu.

Command form

An example is ADB. See the list in Variations and options.

Calling for one of the activities

Key in one of the commands listed in Variations and options and press <ENTER>.

LOADING OVERLAY appears. When successfully loaded, the first display for the selected activity appears. If needed volumes or directories are not found on removable media, prompts appear requesting them. Use the Modify Default Volume Pathnames display to set path to removable media.

Variations and options

The following are the Engineering activities and the commands that call for them:

Table 11-2 Commands and Activities

Eng. Activity	Command
Area Database	ADB
Application Module	AM
Area Names	AREA
Button Configuration	BC
CL Programs	CL, CLK, LK, UNLK
Computing Module	CM
DEB Commands	CMD
DEB EC Command	DEBEC
Console Names	CONS

Eng. Activity	Command
Free Format Logs	FFL
Find Names	FN
Hiway Gateway	HG
HM History Groups	HMG
Logic Blocks	LB
LCN Nodes	LCN
Load Multiple Nodes	LD, LN
Network Interface Module	NIM
Picture Editor	PE
System Maint.	SM
System Wide Values	SWV
Translators*	TR
Unit Names	UNIT
Volume Configuration	VC

*Use the Translators (TR) command only when the software upgrade instructions in the Customer Release Guide tell you to do so.

Additional overlays may be available as custom displays. To see a Help display with custom displays and subcommands, press <HELP>, then <PAG FWD> to page 3 of the Help display.

11.4 Documentation Tool

Purpose

Searches and retrieves data from the on-line databases of LCN nodes and process-connected devices including AMs, CMs, the UCN, and Data Hiway devices. Once data is retrieved, commands within the Documentation Tool can manipulate data to create more customized information. Results of this data manipulation can be sent to a file or to a printer.

Command form

DT

After you key in this command and press <ENTER>, the Documentation Tool overlay is loaded and the initial Documentation Tool display appears.

For instructions for use of the Documentation Tool, see *Documentation Tool Manual*.

Variations and options

The following commands are available from the Documentation Tool's initial display:

FIND— Locates the next occurrence of a pattern, or specified field within the current display.

FILTER— Displays a subset of the data currently on the screen.

SORT— Arranges records within a file in ascending order.

QUERY— Retrieves user-created entity data from the online database.

OUTPUT— Prints or saves the results of previous Documentation Tool operations such as queries and sorts.

DEFINE CNTRL FILES— Creates the directory that stores previously built queries.

OPEN— Displays a file as defined by the pathname entered in the entry ports.

CLOSE— Removes the display from the display stack.

11.5 End Command Processor Session

Purpose

Returns from the Command Processor display to the Engineering Main Menu. If a printer is in use, printing stops and the printer is released. If a file is open, it is closed.

Command form

END

11.6 Equipment List Builder

Reference

For more information about the Equipment List Builder (ELB) command, see the *Equipment List Reference Manual*.

11.7 Execute Command

Purpose

Executes commands found in user-created Command Files, thus providing a way to execute a predefined sequence of commands to perform complex tasks that are frequently accomplished. As the commands are executed, they and any prompts they cause appear on the screen.

Command form

EC NET>VDIR>COMMANDS . EC

Alternate command names are EX and EXECUTE.

The COMMANDS field can contain any of the Command Processor commands and any of the commands in 12—Decision-Making Commands.

Valid Device ID forms are NET and \$Fn.

Use the Text Editor to create your command files, and to enter data in them.

General information can be found in *Text Editor Operation Manual*.

Variations and options

A command file can include one or more of the following directives:

```
EC NET>VDIR>COMMANDS.EC PAR1 PAR2 PAR3
```

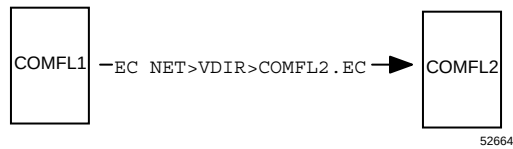
where

PAR1 PAR2 PAR3— Parameter Values

- `&` This is a comment—"&" followed by a space specifies a comment line, which is ignored by the Execute Command.
- `&A`—"&" followed by an "A" causes the Command Processor to enter the Attached Mode. In the Attached mode, prompts generated by the commands must be answered by data in the command file.
- `&D`—"&" followed by a "D" causes the Command Processor to enter the Detached Mode. In the Detached Mode, a prompt generated by a command causes execution to pause until the prompt is answered by someone at the Universal Station. If the Attached mode has not been called for by `&A`, the Command Processor is in the Detached Mode.
- `&F` or `&FILE`—"&" followed by an "F" or "FILE" causes the Command Processor to open a text file. The file is closed when the next `&F` or `&FILE` command occurs, or an end-of-file is encountered or the EC ends.
- `&P`—"&" followed by a "P" causes the text that follows to be sent to the screen and to any printer or file specified by the preceding Prompt Out Command.

Any of the Command Processor commands and any of the Decision- Making Commands can be included in a Command File, including EC. In a Command File executed by an EC Command in another Command File, the `&A` and `&D` directives affect only the file that contains them.

Reference: 12—Decision-Making Commands

**ATTENTION**

In earlier releases, when you put an Engineering Main Menu command (see the list following) in an EC file, the engineering activity is called and the Engineering Main Menu appears. This leaves the remainder of the EC file unexecuted and some memory is not released.

In R400 and later releases, the EC file processor intercepts a main menu command, displays an error message, and continues execution of the EC file.

Commands that call an Engineering Main Menu activity**Table 11-3 Main Menu Commands**

ADB	Area Database
AM	Application Module
AREA	Area Names
BC	Button Configuration
CM	Computing Module
CMD/DEB	Builder Commands
CONS	Console Names
DT	Documentation Tool
FFL	Free Format Logs
HG	Hiway Gateway
HMG	HM History Groups
LB	Logic Blocks
LCN	LCN Nodes
NIM	Network Interface Module
PE	Picture Editor
SM	System Maintenance
SWV	System Wide Values
UNIT	Unit Names
VC	Volume Configuration

Command file examples

The following is an example of a simple command file. Each comment explains what the next command line does.

```
& The following command line directs command lines and
& prompters to file SESSION.EC, as well as to the screen.
PO NET>CL>SESSION.EC
& The next line sends a message to the screen.
&P READY?
& The next command causes execution to pause until
& the operator presses ENTER.
PAUSE
LS NET>ABC>*. * -D
&P THE CATALOG FOR VOLUME ABC IS
&P STORED IN FILE:  SESSION.EC.
```

The following is a command file that uses parameter values. When the file executes, "&1" and "&2" are replaced by parameter values 1 and 2 from the command line.

```
&P Copying &1 and &2

&P Copying &1 and &2

&P &1 Copied to &2
```

The following is a command line to execute the command file, preceding:

```
EC NET>VOL1>CMDFILE.EC NET>VOL1>JOE.EC NET>VOL2>=
where
```

NET>VOL1>JOE.EC– Parameter Value 1

NET>VOL2>=– Parameter Value 2

When executed, this command line and file copies JOE.EC from VOL1 to VOL2. The last message on the screen says:

```
NET>VOL1>JOE.XX Copied to NET>VOL2>=
```

The following example depicts how an EC can be used to back up files. In this example, you may do a backup from any source path to any destination path. The EC asks several questions that you must answer to determine where the files are located and which suffixes they have. The EC also requires the file time and date in order to allow selective file backup; for example, performing backups for a week or an entire month. This command file uses several of the commands described in section 13, Decision- Making Commands.

In the following example the command file,

- gathers the source and destination pathnames,
- uses an LS command to list the desired files,
- writes them to a temporary file named FILES.LT,
- opens file FILES.LT with the &FILE directive,
- reads each line in the file (the third and fourth fields are the file names and suffixes, respectively),
- uses ^R3 and ^R4 to copy the file from the source pathname to the destination pathname,
- and repeats this process for each file in FILES.LT.

```
&P**Begin Backup of Files**
&P

&IF EXISTS FILES.LT & THEN DL FILES.LT

& Q2 "Are you using a Cartridge in the backup? (y or n)"
&IF NOT ^Q2 & THEN &G AGAIN
&Q3 "Do you want to format your Cartridge? (y or n)"
&IF NOT ^Q3 & THEN &G AGAIN
&V9 "Enter the device where the Cartridge is located:"
&V10 "Enter the volume name to initialize the Cartridge with:"

&P
&P The Cartridge in device ^V9 will be formatted and initialized
&P and everything currently on the Cartridge will be lost.
&P
&Q4 "Do you wish to continue to format and initialize the Cartridge? (y or
n)"
&IF NOT ^Q4 & THEN &G AGAIN
CR ^V9>^V10 -FMT -MF 2000

&L AGAIN

&V1 "Enter the extension of the files you wish to backup (e.g. CL, *, EB)."
```

```
&V2 "Enter the device from where the files should be read:"
&V3 "Enter the volume from where the files should be read."
&Q5 "Do you wish to specify a date range to choose the files? (y or n)"
&IF NOT ^Q5 & THEN &G NORANG
&V7 "All ^V1's will be backed up, including and after this date
(mm/dd/yy):"
```

```
&V8 "All ^V1's will be backed up, prior to this date (mm/dd/yy):"
&L NORANG
&P
DO FILES.LT
&IF NOT ^Q5 & THEN &G LSTWO
LS ^V2>^V3>*.^V1 -FROM ^V7 -PRIOR ^V8 -NOHDG
&G LSONE
&L LSTWO
LS ^V2>^V3>*.^V1 -NOHDG
&L LSONE
DO
&Q1 "Do you wish to backup any more files? (y or n)"
&IF ^Q1 &THEN &G AGAIN

&V4 "Enter the device where the files should be copied to:"
&V5 "Enter the volume where the files should be copied to."
&IF EXISTS ^V4>^V5 &THEN &G OKPATH
&P
&P The designated destination does not exist, hit <ENTER> to continue
&P and it will be created for you.
PAUSE
&V6 "Enter the main volume name under the destination device:"
CD ^V4>^V6> ^V5

&L OKPATH
&FILE FILES.LT
&L NEXT
&IF NOT READ &THEN &G DONE
CP ^V2>^V3>^R3.^R4 ^V4>^V5>=
&G NEXT

&L DONE
&P
&P ** Completed Backup of Files**
```

The following shows what appears on the screen as the backup command file, described above, executes:

```
EC EXAMPLE.EC
  **BEGIN BACKUP OF FILES**

ARE YOU USING A CARTRIDGE IN THE BACKUP? (Y OR N)
Y
ENTER THE DEVICE WHERE THE CARTRIDGE IS LOCATED:
$F6
ENTER THE VOLUME NAME TO INITIALIZE THE CARTRIDGE WITH:
EC2
THE CARTRIDGE IN DEVICE $F6 WILL BE FORMATTED AND INITIALIZED AND
EVERYTHING CURRENTLY ON THE CARTRIDGE WILL BE LOST. DO YOU WISH TO CONTINUE
TO FORMAT AND INITIALIZE THE CARTRIDGE? (Y OR N)
Y
CR $F6>EC2 -FMT -MF 2000
Create Volume Complete
ENTER THE EXTENSION OF THE FILES YOU WISH TO BACKUP (E.G. CL, *, EB):
ENTER THE DEVICE FROM WHERE THE FILES SHOULD BE READ:
$F7
ENTER THE VOLUME FROM WHERE THE FILES SHOULD BE READ:
EC
DO YOU WISH TO SPECIFY A DATE RANGE TO CHOOSE THE FILES? (Y OR N)
N

DO FILES.LT
Data Out Complete
```

```
LS &F7>EC>*.XX -NOHDG
```

The list called for by the LS command appears here.

```
Catalog Complete
DO
Data Out Complete
DO YOU WISH TO BACKUP ANY MORE FILES? (Y OR N)
N
ENTER THE DEVICE WHERE THE FILES SHOULD BE COPIED TO:
$F6
ENTER THE VOLUME WHERE THE FILES SHOULD BE COPIED TO:
EC2
CP $F7>EC>ATTACH.EC $F6>EC2>=
Copy Complete
CP $F7>EC>BLANKS.XX $F6>EC2>=
Copy Complete
CP $F7>EC>BLANKS.XX $F6>EC2>=
Copy Complete
CP $F7>EC>EMPTY.XX $F6>EC2>=
Copy Complete
CP $F7>EC>F1.XX $F6>EC2>=
Copy Complete
CP $F7>EC>F2.XX $F6>EC2>=
Copy Complete
CP $F7>EC>F3.XX $F6>EC2>=
Copy Complete
CP $F7>EC>FILE.XX $F6>EC2>=
Copy Complete
CP $F7>EC>FILE1.XX $F6>EC2>=
Copy Complete
CP $F7>EC>FILES.XX $F6>EC2>=
Copy Complete
CP $F7>EC>FILES1.XX $F6>EC2=
Copy Complete
CP $F7>EC>LONGLINE.XX $F6>EC2>=
CP $F7>EC>SUB_STR.XX $F6>EC2>=
Copy Complete

**COMPLETED BACKUP OF FILES**
EC Complete
```

11.8 Data Entity Builder Execute Command (DEBEC)

Purpose

The DEBEC command executes a Data Entity Builder (DEB) EC file. It calls the DEB overlay and then executes the EC file that is specified on the command line. Upon completion, the DEB returns to the Command Processor.

DEBEC does the following:

- Invokes a DEB EC
- Command arguments pass DEB EC file name to DEB and set Overwrite switch (if option is used) in DEB
- Executes from Command Processor (CP) command line or a CP EC file

Command form

```
DEBEC <DEB EC file path> [-OV]
```

where -OV is the Overwrite option

Characteristics

The DEBEC command:

- Cannot be scheduled to run in background mode.
- Is not restricted to the Engineer keylock position; however, keylock requirements for commands within the DEB EC file remain unchanged.

Message at completion

Upon completion of the DEB EC file, you are returned to the Command Processor EC file. The CP display is restored and any error information passed back from the DEB is appended to the CP display.

Error messages

Error messages are displayed upon returning from the DEB to the Command Processor EC file. Execution of the DEBEC command may return any of the following error messages. Refer to Table 10-4.

Table 11-4 Possible DEBEC Error Messages

Error Message	Meaning
(1) Invalid Path OR File Not Found	Unable to access the file path.
(2) SEE ERROR FILE: <filepath>	Error files were generated by the DEB.
(1) Invalid Switch	Switch opted for, but is neither “-ov” nor “-OV.”
(2) INVALID PATHNAME SUFFIX	File path exists, but suffix is not WC.

(1) Generated in the Command Processor

(2) Generated in Data Entity Builder

**CAUTION**

Be careful of DEB EC file sequences, which cause error file name collisions, because error files may be lost.

Error conditions

Two of the resources that may be totally expended by the DEB, depending upon the commands given to it, are:

- Memory resident volume file space
- Heap

Error files in the DEB are created as memory resident files and then copied to their final destination. Very large error files can use up all the available memory resident volume resource. Altering parameters uses significant amounts of heap and altering a large number of parameters for a large number of entities can use up the entire heap.

Backup and restore commands

With TPN R682, you can take a backup of box parameters to a file and later use the file for restoring the values. The backup and restore option is available in xPM, SM and LM. This is also applicable for APM and HPM simulator box parameters.

The backup and restore option is available for the following box parameter type:

- STRINGS
- NUMERICS
- FLAGS
- TIMES

The maximum limit for box parameters that can be configured is as follows:

Parameter type	PM	APM	HPM	SM	LM
Flags	1023	4095	4095	2000	1024
Numerics	2047	4095	4095	1000	1024
Times	NA	4095	4095	NA	NA
Strings	NA	4095	4095	NA	NA

From DEBEC, invoke the backup option using the command “BPB” and invoke the restore option using the command “BPR”.

An example of DEBEC command for backup:

```
V1=NET>RRJ
```

```
BPB V1>ABC.XX 21 03
```

Where ABC.XX is the pathname for backup

V1 represents a volume/directory designation

21 is the box number,

03 is the UCN number

An example DEBEC command for restore:

```
BPR NET>AP>ABC.XX 21 03
```

Where NET>AP>ABC.XX is the pathname for restore

21 is the box number,

03 is the UCN number

The errors that occur during the backup/restore operation are recorded in an error file with .EF extension. The error file name is the same as the .EC user file and is created in the same location as the backup/restore file.

11.9 Floppy (Removable Media) Copy

Purpose

Copies the full content of one removable media to another. The Source and destination media must be mounted in drives that are directly connected to the Universal Station.



ATTENTION

Removable media containing test programs HVTS, CMTS, and LCN Communication Test cannot be copied using Floppy Copy. If you want to copy these programs, HVTS contains a copy command that will let you do so.

Command form

FCP \$F1 \$F2

where

\$F1– Source

\$F2– Destination

Alternate forms of the command name are FCOP and FCOPY.

Message at completion

Media Copy Complete

11.10 Load Multiple Nodes

Purpose

Lets you load one or more nodes by executing a single command, without the need for someone to remain at the console while several nodes are loaded.

Uses the default volumes/directories as sources for the personality images and checkpoint data. These volumes and directories must be on an HM.



ATTENTION

The nodes to be loaded must be shut down or reset before the Load Multiple Nodes command is executed (status must be PWR_ON or QUALIFIED). Shut them down through the appropriate status display.

Command form

```
LD 35 36 38 40
```

where

35 36 38 40 – Node numbers for each node to be loaded separated by at least one space.

Messages output as Load Multiple Nodes executes

As each node is loaded, progress messages like these appear:

```
LOADING NODE 35
```

```
LOADING NODE 36
```

```
LOADING NODE 38
```

```
LOADING NODE 40
```

```
LD COMPLETE
```

Error messages

If the command cannot be executed or if any node can't be loaded, a message appears to explain why. Here are two examples:

```
NODE IS NOT IN PROPER STATE-SHUTDOWN NODE BEFORE  
ATTEMPTING LOAD—The nodes to be loaded must be shut down or reset before  
the Load Multiple Nodes command is executed (status must be PWR_ON or  
READY). Shut them down through the appropriate status display.
```

WRONG NODE TYPE 43—History Modules cannot be loaded with the Load Multiple Nodes command.

For more information about problems with loading nodes, see the *Engineer's Reference Manual*.

Variations and options

Unless you specify otherwise, the LD command loads the on-process personality in each node. You can specify that a US be loaded with a personality other than the default (on-process) personality by adding the abbreviation (OP, UP, or AM) to the node number as in this example:

```
LD 35 UP36 38 AM40 OP37
```

where

UP36— Loads the Universal Personality to node 36.

AM40— Loads the Application Module to node 40.

OP37— Loads the Operator Personality to node 37.

To stop a loading sequence initiated by the LD command, press <BREAK> (hold <CTL> and press <COMMND>). The sequence stops after loading of the current node is complete and before loading of the next node begins.

To see a **help display** for the LD command, key in **LD** (with no node numbers) and press <ENTER>.

11.11 **Pause**

Purpose

The Pause command is used to pause in the Execute Command (EC) to wait for an operator action. For example, it would allow the operator to perform a <CONTROL/BREAK> key combination to abort an execution in progress.

Command form

PS

Enter the Pause command at the Command Processor to pause the US activity, press the <Enter> key to resume activity; you can also press the <Control/Break> key combination to resume activity.

When you place a Pause command in the Execute Command, the system pauses, waiting for an operator action.

Reference

The use of the Pause command with EC is described on one of the TotalPlant Solution's (TPS) on-line Text Editor Help screens. From the Text Editor, press the Help key on the Engineer's keyboard.

11.12 Protect/Unprotect Zip Disk

Purpose

Protects (PROTV) the Zip Disk mounted in the console from being overwritten or deleted. Also cancels such protection (UNPTV).

The Protect and Unprotect commands serve the same purpose as the physical protection for floppies and cartridges; however, this feature is provided as a software, instead of a hardware option for Zip disks.

Command form/Protect

PROTV (\$F1-\$F20)

where

\$F1-\$F20

= Source Zip Drive

Command form/Unprotect

UNPTV (\$F1-\$F20)

Message at completion

After the volume is cataloged to the screen, type "Y" if it is the correct volume. The completion message will then be one of the following:

Media Protect Complete

-or if Unprotect command was used

Media Unprotect Complete

If the command fails or is aborted by the operator, one of the following messages is displayed:

Unable to Protect Media

-or if Unprotect command was used

Unable to Unprotect Media

No query option

```
PROTV ($F1-$F20) -NQ
```

If you include the no query option (-NQ or -NO_QUERY) after the name of the Zip drive, the command will not catalog the volume to the screen nor expect the “Y” verification before completing the command.

EC files use no query option

The no query (-NQ or -NO_QUERY) option should always be used when the PROTV or UNPTV command is used in an EC file. This avoids hanging the EC in a wait for operator verification.

11.13 Pre-Defined Documentation Tool Query

Purpose

This functionality provides the ability to trigger and cancel a pre-defined Documentation Tool query using Command Processor commands, and have the query result output to a file or printer automatically. Additionally, this function enables the user to sort the result by Entity name or by user specified first field. User specified first field is defined as the first field in the “Parameter(s) To Show” list in the pre-built Query.

Three commands are available: QFILE, QPRINT, and QCANCEL. They can also be executed in Scheduled ECs, both in the foreground and background mode. For use in CL programs, refer to the *Engineer’s Reference Manual*.



ATTENTION

The AM CL commands – QFILE, QPRINT, and QCANCEL – cannot be used for invoking prebuilt queries with historized journal access.

Command format

```
QFILE <Descriptor> <Pathname> <XX>
QPRINT < Descriptor> $P<n> <XX>
QCANCEL
```

Where,

QFILE is the keyword for specifying a query whose result should be output to a file.
QPRINT is the keyword for specifying a query whose result should be output to a printer.

QCANCEL is the keyword for canceling the running query being executed by a Doc Tool background task.

<Descriptor> is a required field and specifies the name of the pre-defined query.

<Pathname> is a required field and specifies the path to the file in which to put the query results.

\$P<n> is a required field and specifies the printer ID on which the query results are to be printed.

<XX> is an optional field which can take the values as XX = [-SORT | -SORTF | -NSORT].

The optional fields provided for sorting are further explained below:

Sort Directive	Sorted By
-SORT	Sort the Query Result by the Entity Name.
-SORTF	Sort the Query Result by the user specified first field.
-NSORT	Explicitly does not sort the Query Result.

Note: Setting the QRYSORT PSDP value to Node Global Type will override all sort operations.

Command examples

To invoke a pre-defined query from the Command Processor, specify the descriptor of the pre-defined query and the pathname or printer number. In the following examples, the pre-defined query “HGPOINTS” is to be executed with the results going to file HGPOINTS.XX in the QFILE command and to printer \$P2 in the QPRINT command.

```
QFILE HGPOINTS NET>TEMP>HGPOINTS
QPRINT HGPOINTS $P2
```

To cancel the query, the Command Processor command is QCANCEL

To invoke the sort query result, any of the optional Sort Directives may be used. To sort the Query Result by point name:

```
QFILE HGPOINTS NET>TEMP>HGPOINTS.XX -SORT
QPRINT HGPOINTS $P2 -SORT
```

To sort the Query Result by user specified first field:

```
QFILE HGPOINTS NET>TEMP>HGPOINTS.XX -SORTF
QPRINT HGPOINTS $P2 -SORTF
```

To not sort the Query Result:

```
QFILE HGPOINTS NET>TEMP>HGPOINTS.XX -NSORT
QPRINT HGPOINTS $P2 -NSORT
```

To sort the results of the pre-built query “HGPOINTS” by entity and print the results to \$P8, issue the following command.

```
QPRINT HGPOINTS $P8 -SORT
```

In order to make a Command-Processor-command-initiated pre-defined query to run on a periodic basis, the following steps need to be taken:

1. Build an EC file with the QFILE or QPRINT command in it.
2. On the Command Processor command line, use the SCHED command to schedule the EC file to be executed, either in a foreground or background mode at the desired periodic rate.

QRYSORT Processor Status Data Point (PSDP)

In all the above examples, the query results depend on the Sort Directive used; however, if the node global sort value is set in the QTYSORT PSDP, it overrides the Sort Directive set in the Command Processor command. All the queries will be executed on the same node where the query is invoked from the Command Processor.

```
$PRSTS<XX>.QRYSORT = <Y>
```

Where,

- XX is the node number of a Universal Station.
- Y is the QRYSORT value from 0 to 6. Explanation of these values (0-6) are as follows:

TYPE	QRYSORT Value	Definition
Temporary	0	“No Sort.” The Sort Directive in the CL/CP query will override this option.
	1	“Sort by Entity Name.” Automatically reset to ‘No Sort’ at completion of Query. The Sort Directive in the CL/CP will override this option.
	2	“Sort by user specified first field.” Automatically reset to ‘No Sort’ at completion of Query. The Sort Directive in the CL/CP will override this option.

TYPE	QRYSORT Value	Definition
Node Global	3	“No Sort.” This option overrides the CL/CP Sort Directive. This value is not automatically reset. Once set, this option can only be changed to another Node Global type or Reset.
	4	“Sort by Entity Name.” This option overrides the CL/CP Sort Directive. This value is not automatically reset. Once set, this option can only be changed to another Node Global type or Reset.
	5	“Sort by first user specified field.” This option overrides the CL/CP Sort Directive. This value is not automatically reset. Once set, this option can only be changed to another Node Global type or Reset.
Reset	6	Used to Reset the Node Global options. Once a Node Global option is set, it can be changed to a temporary type only by setting this option. Setting this option, resets the PSDP value to 0.

- Refer to the Engineering Reference Manual’s discussion of CL Predefined Documentation Tool Queries for more information on this topic.

11.14 Schedule EC Execution (R530)

Basic Information

R530 provides the ability to schedule EC execution through two new commands, Schedule and Unschedule. These new commands reside in the Command Processor. ECs can be scheduled to run in background or foreground mode.

In background mode, the EC runs “invisibly” while other UNP functions are being performed. All output (normally directed to the screen) is instead directed to a “Background Results” file called (a BR file). The BR path and filename are the same as the EC path and filename, and the extension, which is new, is “.BR”.

In background mode, only a subset of the command processor commands is allowed (refer to the following table for a list of background commands). Any command not allowed in the background mode will cause the EC to abort if encountered. Also, the background mode maintains its own default path so ECs running in background mode should “set path” accordingly.

In foreground mode, the EC runs just as if it had been manually initiated. The Universal Station must be at the Main Menu or at the Command Processor for the foreground EC to start, and the Command Processor input port must be blank.

ECs can be scheduled to run Hourly, Daily, Weekly, Monthly, or Once. The user may supply the initial start date and time. If not supplied, the initial date and time defaults to current date and time.

On startup of node “PN,” ECs are scheduled as defined in system file “Sched_PN.SU” on the NET>&ASY volume. The absence of such a file indicates that there are no ECs scheduled on the node. This file is automatically created and maintained by the execution of Schedule and Unschedule commands.

Example

The weekly backups are dependant on an engineer initiating a backup. R530 provides the ability to schedule EC execution.

At US12, the engineer would type the following command one time:

```
SCHED -B -W 06 OCT 96 23:59 NET>USR1>BKUP.EC
```

This will schedule BKUP.EC to run in Background mode, on node 12, Weekly, beginning on the 6th of October at 23:59. Because the timeframe is Weekly, and 06 OCT 96 is a Sunday, the EC then executes every Sunday at 23:59. The EC output is captured in file NET>USR1>BKUP.BR.

New Status Account Object

A new Status Accountant object on the Status Detail Display will track the node startup scheduling and execution of background ECs.

Background Commands

The following table provides a list of background Commands.

Table 11-5 Background Commands

COMMAND:	Format:	Notes:
Change Volume Desc.	SVD	A
Copy File	CP	A,B
Copy Volume	CPV	A,B
Create Directory	CD	A
Create Volume	CR	A
DataOut (file)	DO dev>vol>file.xx	A,C
DataOut (printer)	DO \$Pn	A,C
Delete Directory	DD	A
Delete File	DL	A
Execute Command File	EC	A
Floppy Copy	FCOPY	A,B
List Volumes Net	LSV	A
Modify Directory Desc.	MFD dev>vol desc	A
Modify File Descriptor	MFD dev>vol>file.xx desc	A
Move File	MV	A,B
Print	PR	A
PromptData (On/Off)	PD ON or OFF	A,
PromptOut (file)	PO dev>vol>file.xx	A,C
PromptOut (printer)	PO \$Pn	A,C
Protect	PROT	A
Protect Volume	PROTV \$Fxx	A,D
Pre-defined Doc Tool Query (file)	QFILE <descriptor> <Pathname> <xx>	A
Pre-defined Doc Tool Query (printer)	QPRINT <descriptor> \$P<n> <xx>	A
Cancel Pre-defined Query in Progress	QCANCEL	A
Rename File	RN dev>vol>file.xx file2	A
Rename Volume	RN \$Fn>vol VOL2	A
Set Path (specific)	SP dev>vol	A
Slew	SLW	A
Status (Device / Volumes)	STA	A
Status Net	STA NET or PN:nn	A
Time	Time	A
Unprotect	UNPT	A
Unprotect Volume	UNPTV \$Fxx	A,D
Write Boot	WB	A
Notes: A. Errors and prompts are noted in the BR file. B. A mount request causes the EC to abort. C. With no argument, output to the specified device or file is halted, but the output continues to be sent to the “.BR” results file. D. The listing and the confirmation are waived.		

11.15 SCHEDULE Command

Purpose

The SCHEDULE command is available from the command processor. It adds the scheduling information for an EC to an internal data structure and it writes the contents of that structure to the SCHED_pn file on disk. The internal structure is then used by the scheduler to determine which ECs to initiate.

The format of the command is SCHEDULE, SCHED, or SCH followed by any switches, timebase, start time, path, and parameters as shown in the following examples. The order in which the switches and parameters are listed is enforced.

Command form

To schedule an EC:

```
SCHED [-B/-F] [-Q/-H/-D/-W/-M] [Date] [Time] Dev>Vol>Filename.EC  
[P1 P2 P3...] [-B Background, -F Foreground] [-O Once, -H  
Hour, -D Day, -W Week, -M Month]
```

Where:

Underline indicates a default and items in [brackets] are optional

P1, P2, P3 are parameters to the EC

Example 1: Background, hourly, current date, and time

```
SCHEDULE -B -H 01 NOV 96 04:00 Net>EC>Bkup.EC $F12 $P2
```

Example 2: Default to background, once and current date and time.

```
SCHED Net>EC>Bkup.EC $F12 $P2
```

To abort the running background EC (Does NOT Unschedule, see 11.20 to Unschedule):

```
SCHED -A
```

To load or save data from the SCHED_pn file on disk:

```
SCHED (-L/-S)
```

To show the status of all ECs currently scheduled (List Status):

```
SCHED -LS
```

To show the commands used to schedule all ECs currently scheduled (List Commands):

```
SCHED -LC
```

To get help on the command:

```
SCHED
```

Schedule Command messages

The following tables list the various formats of the SCHEDULE command and the possible messages associated with them.

Table 11-6 SCHED (Schedule an EC)

Schedule Command	Message
Invalid Command Format	Invalid parameter order or miscellaneous error.
Invalid Command Switch	Invalid switch or switch combination.
Invalid Time Format	Invalid date or time or invalid order of date and time.
Invalid Path Or File Not Found	Invalid path or file does not exist.
Cannot Schedule More Than Max.	The maximum number of scheduled ECs has been reached. The EC has not been scheduled.
EC Scheduling Complete	Scheduling completed without errors.
Cannot Access Sched_xx.SU File	The &ASY startup file could not be accessed to record the newly scheduled EC. The internal structure and the startup file are now "out of synchronization."
SCHED_EC DISABLED UNTIL RELOAD	The SCHEDULE EC function has encountered an illogical condition and it has been disabled. No further schedule commands are allowed and no further scheduled EC execution is performed.

Table 11-7 SCHED -A (Abort the EC running in background mode)

Schedule Command	Message
Background Abort Initiated	The abort command has been sent to the background task. It may take some time for the background EC to abort.
No Background EC Active	There is no EC currently running in background.
SCHED_EC DISABLED UNTIL RELOAD	Defined earlier.

Table 11-8 SCHED -S (Save)

Schedule Command	Message
Save To Sched_pn.SU Complete	The save completed without errors.
Cannot Access Sched_pn.SU File	The startup file on &ASY could not be accessed to read or write data. The internal structure may or may not have been updated but its status was defined in a separate message.
SCHED_EC DISABLED UNTIL RELOAD	Defined earlier.

Table 11-9 SCHED -L (Load)

Schedule Command	Message
EC Scheduling Load Error	The load was not initiated. The startup file is empty or unavailable, or there was an internal error.
EC Scheduling Load Complete	The load completed without errors.
Cannot Access Sched_pn.SU File	Defined earlier.
SCHED_EC DISABLED UNTIL RELOAD	Defined earlier.

Table 11-10 SCHED -LS &SCHED -LC (List Status and List Commands)

Schedule Command	Message
There are No ECs Scheduled	
SCHED_EC DISABLED UNTIL RELOAD	Defined earlier.

11.16 Sequence Program Loading Command

Purpose

With R640, a Sequence Program Loading command is available from the Command Processor and an EC file.

Using this command, the user can:

- Load a sequence program into a PM, APM, or HPM process module point,
- Start or stop the sequence program,
- Change the module operating status to off, or
- Change the mode of the sequence program.

Before attempting to load a sequence program, ensure that the compiled object of the program is loaded into the appropriate History Module directory.

Command Form

The format of the command is SEQCMD or SC followed by the optional switches that are listed below. The user should use at least one option switch with this command.

```
SEQCMD <point name> <[-stop/-stopwo] | [-off] | [-load  
<sequence program name>] | [-auto / - semi / -sngl] | [-  
start]>
```

Where,

`point name` – name of the process module data point.

`stop` - stops the execution of the sequence program. If the stop option fails, the command is aborted.

`stopwo` (stop with override) – stops the execution of the sequence program but when the option fails, the command is not aborted. However, the command is aborted if the option failures are encountered after “stopwo”.

`off` - sets the module operating state to OFF

`load <sequence program name>` - loads a sequence program into the specified process module point. OPER should be in the OFF state before loading the sequence, otherwise an error will occur.

`auto` – changes the execution mode of the sequence to automatic

`semi` – changes the execution mode of the sequence to semi-automatic

`sngl` - changes the execution mode of the sequence to single step

start - starts the sequence program

The following table defines the valid command-processor sequence program commands.

Table 11-11 Sequence Program Command and Definitions

Command	Definition
SEQCMD <point name> -stop	Stops the sequence when the execution state is in RUN in the specified process module.
SEQCMD <point name> -stopwo	Stops the sequence when the execution state is in either RUN or NL.
SEQCMD <point name> -stop -off	Changes the operating state to OFF when the execution state is in RUN.
SEQCMD <point name> -stopwo -off	Changes the operating state to OFF when the execution state is in either RUN or NL.
SEQCMD <point name> -off	Changes the operating state to OFF when the execution state is not in RUN.
SEQCMD <point name> -stop -off -load <sequence name>	Loads a sequence when the execution state is in RUN.
SEQCMD <point name> -stopwo -off -load <sequence name>	Loads a sequence when the execution state is in either RUN or NL.
SEQCMD <point name> -off -load <sequence name>	Loads a sequence when the execution state is not in RUN.
SEQCMD <point name> -load <sequence name>	Loads a sequence when operating state is in OFF state.
SEQCMD <point name> -stop -off -load <sequence name> -start	Loads and starts a sequence when execution state is in RUN. The sequence will be started in default execution mode (AUTO).
SEQCMD <point name> -stopwo -off -load <sequence name> -start	Loads and starts a sequence when execution state is in either RUN or NL state. The sequence will be started in default execution mode (AUTO).
SEQCMD <point name> -off -load <sequence name> -start	Loads and starts when the execution state is not in RUN and operating state is not in OFF. The sequence will be started in default execution mode (AUTO).
SEQCMD <point name> -load <sequence name> -start	Loads and starts when the operating state is in OFF. The sequence will be started in default execution mode (AUTO).
SEQCMD <point name> -start	Starts a sequence when execution state is in LOAD, PAUSE, FXXX and EXXX. The sequence will be started in default execution mode (AUTO).
SEQCMD <point name> -stop -off -load <sequence name> -auto -start	Loads and starts a sequence when execution state is in RUN. The sequence will be started in automatic (AUTO) execution mode.

Command	Definition
SEQCMD <point name> -stopwo -off -load <sequence name> -auto -start	Loads and starts a sequence when execution state is in either RUN or NL. The sequence will be started in automatic (AUTO) execution mode.
SEQCMD <point name> -stop -off -load <sequence name> -semi -start	Loads and starts a sequence when execution state is in RUN. The sequence will be started in semi-auto execution mode.
SEQCMD <point name> -stopwo -off -load <sequence name> -semi -start	Loads and starts a sequence when execution state is in either RUN or NL. The sequence will be started in semi-auto execution mode.
SEQCMD <point name> -stop -off -load <sequence name> -sngl -start	Loads and starts a sequence when execution state is in RUN. The sequence will be started in single execution mode.
SEQCMD <point name> -stopwo -off -load <sequence name> -sngl -start	Loads and starts a sequence when execution state is in either RUN or NL. The sequence will be started in single execution mode.

Sequence Command Messages

The following table lists the messages that can be displayed depending on the option that is used with SEQCMD.

Table 11-12 SEQCMD Option Messages

Option	Message at successful completion	Error Message
-stop	SEQ. <seq. name> ON PT. <point name> STOPPED.	SEQ. <seq. name> ON PT. <point name> COULD NOT BE STOPPED.
-stopwo	SEQ. <seq. name> ON PT. <point name> STOPPED.	SEQ. <seq.name> ON PT. <point name> COULD NOT BE STOPPED BUT OVERRIDEN.
-off	SEQ. <seq. name> ON PT. <point name> SET TO OFF.	PT. <point name> COULD NOT BE SET TO OFF.
-load	SEQ. <seq. name> ON PT. <point name> LOADED.	SEQ. <seq. name> ON PT. <point name> COULD NOT BE LOADED.
-mode	SEQ. <seq. name> ON PT. <point name> SET TO <new mode>.	MODE OF SEQ. <sequence name> ON PT. <point name> COULD NOT BE CHANGED.
-start	SEQ. <seq.name> ON PT. <point name> STARTED.	SEQ. <seq.name> ON PT. <point name> COULD NOT BE STARTED.

Loading Sequence Program from EC file

You can also use the SEQCMD and the associated options in an EC file. You can start the EC file from a command prompt or schedule the EC file in the foreground. The SEQCMD command cannot be executed in the background mode.

The following is an example of an EC file that contains the SEQCMD command:

```
--Start of EC file
--It is assumed that the sequence is in RUN state (EXEC = RUN)
  SEQCMD 6402PM01 -stop -off -load SEQ1 -auto -start
  SEQCMD 6402PM02 -stop -off -load SEQ3 -auto -start
--End of EC file
```

Command Form

The EC file can be run from the command processor issuing the following command:

```
EC 6402CMD.EC
```

The EC file can be executed in the foreground by the following command:

```
SCHED -F 22 NOV 02 22:00 NET>TEST>6402CMD.EC -O
```

Where,

- o- executes once at a given date and time
- h- hourly
- d- daily
- w-weekly
- m-monthly

The EC file is executed in the foreground by using the -o option. The periodicity can be defined by using the options like, -h, -d, -w, or -m.

To exit from the execution of the EC file, press <CTL>and <BREAK> keys.

Table 10-13 SEQCMD Error Messages

Error Message	Definition
FILE NOT FOUND (19)	Misspelled SEQCMD name.
See Help for more help on this command	SEQCMD does not specify process module point name and option switches.
Incomplete command	SEQCMD without either process module point name or option switches.
Incomplete command	The -load switch is used without sequence name.
Sequence name not found	Sequence name not in the NIM library.
Invalid Entity Id	Invalid process module point name.
Point is not a procmo point	Point specified is not a process module point.
Invalid option	An option switch is misspelled.
OPER not in OFF State SEQ. <sequence name> ON PT. <point name> COULD NOT BE LOADED	Load operation invoked and OPER not in OFF state.
Insufficient slot size SEQ. <sequence name> ON PT. <point name> COULD NOT BE LOADED	Load operation fails due to insufficient process module slot size.
More than one mode option	More than one mode option used.
Sequence Operation Failed <Operation specific messages>	SEQCMD fails due to any sequence related errors.
The error log file will have following error message "illegal background command"	SEQCMD is used in an EC file scheduled in background.
Incomplete command	When the point name is invalid and also no options are passed in the command.
Sequence Operation Failed SEQ. <seq. name> on PT. <point name>could not be stopped	-stop option fails

Miscellaneous Commands – Sequence Program Loading Command

Error Message	Definition
Sequence Operation Failed SEQ. <seq. name> on PT. <point name> could not be STOPPED but overridden	-stopwo option fails
Sequence Operation Failed PT. <point name> could not be set to OFF	-off option fails
<load fail specific message> SEQ. <sequence name> on PT. <point name> could not be LOADED	-load option fails
Sequence Operation Failed MODE of SEQ <sequence name> on PT. <point name> could not be changed	-auto/semi/sngl option fails
SEQ <seq. name> on PT. <point name> could not be STARTED	-start option fails
Sequence name mismatch	When the sequence name is not associated with the point name.
Sequence not loaded on the point <option specific failure message>	When stop/stopwo, mode change and start option are executed on a point with sequence not loaded in it.
NODE busy	Load option is executed when console manager is busy.
HIWAY COMMUNICATION ERR	Load operation initiated when the box is in ALIVE state or controller personality is being loaded.
Sequence Name mismatch	Load operation initiated while restoring.
Invalid Entity ID	Any operation when NIM is switched OFF.
Insufficient Slot Size SEQ. <sequence name> ON PT. <point name> COULD NOT BE LOADED	Load operation initiated on a process module point that has an insufficient slot size.
Incomplete Command	Sequence name starts with "-" (hyphen)
Invalid Option	Space after "-" in the option. Example: SC <point name> - "<space>" - off

11.17 Set User Pathname

Purpose

Specifies the pathname for the user volume/directory currently in use. This has the same effect as modifying the [USER DEFLT PATH] port on the Modify Default Volume Pathnames display.

Once you specify the user volume/directory pathname, you need to enter only the file name and its suffix for most operations that would otherwise require a full pathname.

Because the SP command can be entered directly from the Command Processor or from the Data Entity Builder through use of the <ESC> key, this command provides a convenient method for changing the user volume/directory pathname without leaving the CL display or the DEB.

Command form

SP NET>VDIR>

Example: SP \$F2>PNTS>

An alternate form of the command name is SET.

Message at completion

Set User Path Complete

11.18 Short Cuts and “Wildcard” Characters

Purpose

Features and functions that can save you time and effort are described here.

Presetting the user pathname

If the user pathname displayed in the upper-right corner of the Command Processor display is correct, you can save work by omitting the device identity and volume/directory-name portions of any command line. You can set it up by modifying the [USER DEFLT PATH] port on the Modify Default Volume Pathnames display. Instructions for doing this are in section 2, Modifying Default Volume Pathnames.

For example, suppose the user pathname appears as follows:

USER PATH : NET>VOL1>

To copy file JOHN.EL to file JANE.EL on the same volume (VOL1), you could key in this short form of the command line:

CP JOHN.EL JANE

When you press <ENTER>, the copy is made just as if you had keyed in the full form of the command line, like this:

CP NET>MYVOL1>JOHN.EL NET>MYVOL1>JANE

Call other Engineering activities

You can call up other Engineering activities directly from the Command Processor display, without returning to the Main Menu. Instructions are provided with in Call Engineering Activities.

Load several nodes at once

You can load one or more nodes directly from the Command Processor display. Instructions are with in Load Multiple Nodes.

Wildcard characters

Wildcard characters "*" and "?" let you save keystrokes, and they let you define operations on multiple files and volumes/directories with the same name and different suffixes, or with the same suffix and different names.

* and ? in command lines have the following meanings:

* stands for one or more characters in this file or suffix field, beginning at this character position, then to the right, to the end of the field. * cannot be used in the eighth (right-most) position in the file field.

? stands for any single character in the file at this location.

The following are examples of the use of the wildcard characters.

Suppose you have the following text files:

ABCDEFGH.EL

ABCJEFGH.XX

ABCKEFGH.YY

PR ABC?EFGH.*—Causes the content of all three of these files to be displayed or printed. Any 8-character file whose name begins with “ABC” and ends with “EFGH” is displayed or printed; all suffixes are accepted.

PR ABC*.EL—Causes the content of file ABCDEFGH.EL and any other “.EL” files that begin with “ABC” to be displayed or printed.

PR *.EL—Causes the content of all files with any file names and with the suffix “EL” to be displayed or printed.

PR ABCDEFGH.?L—Causes the content of ABCDEFGH.EL and any other ABCDEFGH files with any 2-character suffix ending in “L” to be displayed or printed.

11.19 **TIME Command**

Purpose

The TIME command outputs the date and time to the current output device in Gregorian format. It has only one format, “TIME,” and it ignores any arguments. This command is added as part of the Schedule EC package to allow users to document ACTUAL run times as opposed to the SCHEDULED run times.

Example:

Embed this command in an EC at line 1.

Schedule the EC to run in background mode at 2:00 P.M.

Because of overruns, the EC does not actually run until 2:40 P.M.

The output from the EC, captured in the “.BR” file, shows the actual run time, 2:40 P.M.

11.20 UNSCHEDULE Command

Purpose

The UNSCHEDULE command is available from the command processor. It removes the scheduling information for an EC from the internal data structure and it writes the contents of that structure to the SCHED_pn startup file on disk. The internal structure is then used by the scheduler to determine which ECs to initiate.

The format of the command is “UNSCHEDULE,” “UNSCHED,” or “UNSCH,” and the functionality is as follows:

To show the UNSCHEDULE help:

```
UNSCHED
```

To unschedule all scheduled ECs:

```
UNSCHED *
```

To unschedule an EC with Identifier “ID#”:

```
UNSCHED ID#
```

e.g. “UNSCHED 5”

To unschedule the range of ID numbers from “X” to “Y” inclusive:

```
UNSCHED X-Y
```

e.g. UNSCHED 5-22

The following table lists the arguments for the UNSCHEDULE Command.

Table 11-13 UNSCHEDULE

Argument	Description	Notes
No arguments	This provides a brief help screen.	A
ID# (A number 1 through 50)	This identifies the EC to unschedule.	B
ID RANGE (for example, 5-22)	This identifies a range of ECs to unschedule.	C
* (asterisk)	This indicates that all ECs should be unscheduled. Note that running ECs will not be unscheduled. They must first be aborted.	
Notes: A. The help information scrolls onto the command processor screen. B. The "ID" number is the index of the internal data structure that specifies which EC to unschedule. IDs can be viewed through the "SCHEDULE -LS" command. C. A single range may be entered. The start and stop indices must be separated by only a "dash" character. A separate message is displayed for each EC unscheduled.		

UNSCHEDULE Command messages

The following table lists the various formats of the Unschedule Command.

Table 11-14 UNSCHEDULE Commands

Unschedule Command	Message
Invalid Command Format	Invalid parameter or miscellaneous error.
Index Not Found	The index provided is not in the schedule.
Unschedule Complete	The unschedule completed.
Unschedule Complete For # xx	For each ID unscheduled in a range.
Cannot Access Sched_PN.SU File	Defined earlier.
Cannot Unschedule A Running EC	An EC must be stopped to be unscheduled.
SCHED_EC DISABLED UNTIL RELOAD	Defined earlier.

11.21 Write Boot

Purpose

Writes a boot record on the specified volume/directory. The boot record is a software routine that allows the hardware and firmware to begin the personality-loading process.

If you are making backup copies of a personality's cartridge or floppies (Univ, Opr, or Maint), after you copy the first removable media in the personality set, you must execute a Write Boot command to that removable media, or it will not be possible to load the personality from that removable media set.

Command form

WB \$F1>&LDR

where

\$F1>&LDR– Write Boot to the Universal Personality volume or directory mounted in removable media drive 1.

WB PN:40>!9np

where

PN:40>!9np– Write Boot to the HM local volume/directory (!9np) in an HM running its initialization personality. NOTE: These forms work because boot files have already been copied to those directories.

9np– “np” is the node-pair number for the HM.

Alternate command names are WRITE_BOOT and WRITE_B.

Message at completion

Write Boot Complete

12. Recovering from Errors

12.1 Overview

Purpose of this section

This section provides guidance recovering from errors that may occur when you are modifying Default Volume Pathnames or when you are executing commands through the Command Processor display.

Section content

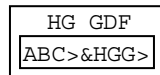
If you are using the Modify Default Volume Pathnames display, see section 12.2, Recovering from Pathname Errors.

If you are using the Command Processor display, see section 12.3, Recovering from Command Errors.

12.2 Recovering from Pathname Errors

Default pathname error indications

One or more of the ports (targets) has red characters or a red outline and an error message appears at the lower-left of the display.

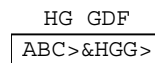


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Default pathname errors and causes

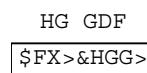
The following are examples of the error messages with the causes of the errors:

- **DEVICE ID SYNTAX ERROR**—The device ID is neither NET nor \$Fn, where n is a removable media drive number. This is an example:



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- **DEVICE NOT FOUND**—The device ID names a device that is not NET and is not that of a removable media drive connected to this console. This is an example:



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- **VOLUME ID SYNTAX ERROR**—The volume name has more than four characters or has one or more illegal characters. This is an example (“&HGG5” has one too many characters):

HG GDF
\$F2>&HGG5>

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Correcting default pathname errors

Use the following procedure:

Table 12-1 Correct Default Volume Pathname Errors

Step	Action	Result
1	Type over wrong characters or press <CLR-ENT> and key in a whole new pathname, then press <ENTER>.	All pathnames should be redisplayed in blue and no ports should have a red outline. If you still have errors, go to step 2.
2	Repeat step 1 until there are no errors.	All pathnames should be redisplayed in blue and no ports should have a red outline.

12.3 Recovering from Command Errors

Command execution error indications

Error indications include the following:

- An error message on the screen.
- Failure to get the expected action or response.
- The “message of completion” (defined for each command in section 3, Command Processor Commands) does not appear.

Identifying the cause of a command error

We suggest you try one or more of the following:

- Determine if the problem is in the command line or is caused by a device malfunction.
- In most cases, you can determine from the error message whether the problem is in your command line or if some device is not functioning properly.
- For help in interpreting these messages see the Messages Directory Manual.
- See the 3—Command Processor Commands section for detailed command and option information.

Correcting command execution errors

Use the following procedure:

Table 12-2 Correct a Command Execution Error

Step	Action	Result
1	Re-enter the command line or key in a new command line and press <ENTER>.	The command is executed and the “complete” message appears.

13. Decision-Making Commands

13.1 Overview

Purpose

In your command files you can use the following commands to ask questions, make decisions, and respond to various conditions.

For more information about command files, see section 11.7, *Execute Command (Miscellaneous Commands)*.

Many of the decision-making commands are used in the .EC files provided by Honeywell, to assist in upgrading LCN-based systems from one software release to another.

Command list

The decision-making commands are as follows:

Table 13-1 Decision-Making Commands

Command	Type	Syntax
&Qn &Vn &IF &THEN	Question Question Directive Directive	&Qn "question" &Vn "question" &IF (active function) &THEN (then clause)
&ELSE &(F)ILE &G &L	Directive Directive Directive [go to (label)] Directive [go to (label)]	&ELSE (else clause) &(F)ILE &G (label) &L (label)
&Q (R)EAD AND OR	Directive [quit] Active function Active function Active function	&Q (Active function) (Active function) (Active function)

Command	Type	Syntax
NOT	Active function	NOT (active function)
EXISTS	Active function	EXISTS (pathname)
^Rn	Active function	^Rn
^Qn	Active function (answer)	^Qn
^Vn	Active function (answer)	^Vn
& space	Comment line	(No print)
&P	Comment line	Display on screen or printer
&A	Active function (answer)	&A prompts answered within command file
&D	Active function (answer)	&D prompts answered from keyboard
&S	Active function (store)	&S &V1 OR &Q1 string

The &, &A, &P, and &D directives may also be useful in .EC files with decision-making commands.

These directives are defined in section 11.7, Execute Command (Miscellaneous Commands).

13.2 Command Descriptions

&Qn: Question, yes or no

“n” is an integer from 1 to 32.

Asks a question and expects a yes or no answer. The answer is stored in the corresponding ^Qn item as a true (yes) or false (no) value that can be acted upon by one of the logical functions (AND, OR, and NOT) or an IF command. A “Y” is recognized as a “yes” answer and an “N” is recognized as a “no” answer. For example,

```
&Q1 "DO YOU HAVE 68000 AM?"

&Q2 "DO YOU HAVE 68020 AM?"

&

&IF ^Q1 &THEN CP &1>&AMO>AMONP?.PI NET>&AMO>= -D

&IF ^Q2 &THEN CP &1>&AMO>AMONPH?.PI NET>&AMO>= - D

&
```

As each question appears on the screen, execution pauses to wait for the answer. Key in Y or N (press <ENTER> after keying Y or N) as the answer. Each answer is stored in ^Q1 and ^Q2 as true (yes) or false (no). The &IF/&THEN command lines test those values to decide if the copy command on the same line should be executed (true) or not (false).

The &Qn question can also be made to act as an active function by including in the &IF line an &ELSE directive after the &THEN directive. For example,

```
&IF ^Q1 &THEN PR NET>HNV1>TST.XX &ELSE PR
NET>HNV1>STS.XX
```

If the user answered with Y, TST.XX is displayed by the PR command, and if the answer was N, STS.XX is displayed.

To use quote marks (") within a question, use any character other than A through Z and 0 through 9 as the delimiters. For example,

```
&Q1 !DO YOU WANT "FAST" UPDATES? !
```

The exclamation points (!) act like the quote marks (") did in the other examples and the question on the screen looks like this:

```
DO YOU WANT "FAST" UPDATES?
```

&Vn: Question, character-string answer

“n” is an integer from 1 to 32.

Asks a question and expects an answer that consists of one or more printable/displayable characters. The answer is stored in the corresponding ^Vn item as a character string that appended to subsequent lines in the command file. For example,

```
&V1 "WHERE DO YOU WANT THIS FILE TO BE STORED?"
```

&

```
CP &1>VOL1>MYFILE.XX ^V1= -D
```

&

When the question appears on the screen, execution pauses to wait for the answer. Key in pathname NET>VOL1> as the answer, and that answer is stored in ^V1. When the copy command line is executed, the content of the ^V1 item (NET>VOL1>) is used as the pathname for the destination of the file to be copied.

&IF: If directive

Tests the true or false value in the active function(s), and execution proceeds as indicated by that value and the &THEN and &ELSE functions on the same command line. The active functions can be nested. “&IF” must be at the beginning of the command line. For example,

```
&IF ^Q1 &THEN CP &1>&AMO>AMONP?.PI NET>&AMO>= -D
```

&

```
&IF ^Q1 AND ^Q2 &THEN &G LABEL1 &ELSE &G LABEL3
```

For the first command line, if the value in ^Q1 is true (answer was “yes”), the copy command on that line is executed. Otherwise, the copy command is skipped and the next command line is evaluated. For the second command line, if both ^Q1 and ^Q2 contain “true,” the &G item transfers control to the line named LABEL1. Otherwise, the &ELSE item transfers control to the line labeled LABEL3.

If ^Qn functions are nested, parentheses must be used. For example,

```
&IF ^Q1 AND (^Q2 OR ^Q3) &THEN &G LABEL1 &ELSE &G  
LABEL3
```

&THEN: Then directive

Defines what is to take place if the active function(s) evaluated by the IF directive is true. The (then clause) defines the action to take and must be on the same line as &THEN. “&THEN (then clause)” must succeed the &IF directive and must precede any &ELSE directive. The (then clause) can be any valid command line or any directive except &L, &IF, &THEN, or &ELSE. For example,

```
&IF ^Q1  
  
&THEN CP &1>&AMO>AMONP?.PI NET>&AMO>= -D  
  
&
```

If the content of ^Q1 is true (answer was yes), the copy-command line is executed. If ^Q1 contains “no,” the copy command is not executed and the next line in the command file is executed. If the (then clause) should be omitted, an &ELSE directive (if present) is ignored, and the next line in the command file is executed.

You could use an “empty” &THEN clause. For example,

```
&IF ^Q1 &THEN &ELSE CP &1>&AMO>AMONP?.PI NET>&AMO>= -D
```

If the user answered Y, there is no action. If the answer was N, the CP command is executed.

&ELSE: Else directive

Specifies that some action, other than execution of the next command line, is to take place. This is an optional directive that, when used, must immediately succeed the &THEN directive. The (else clause) can be any valid command line or any directive except &L, &IF, &THEN, or &ELSE. For example,

```
&IF ^Q1 &THEN &G LABEL1 &ELSE &G LABEL3  
  
&
```

If ^Q1 doesn't contain “true,” control is transferred to the command line that follows the line labeled LINE3.

**ATTENTION**

.EC files can call for other .EC files that include &THEN or &ELSE clauses. If they do, the range of any &Q, &G, &L, &IF, &A, and &D commands are strictly within the range of the file in which they occur.

&F(ILE): File directive

Opens a file using the default pathname unless another is specified. For example,

```
&FILE $F1>ETA>JOHN.XX
```

The file is closed when the next &F(ILE) occurs, or when the .EC ends.

&G: Go-to function

Specifies that control is to be transferred to the command-file line immediately after the specified label location. The label location must be defined by one &L (label) item somewhere in the command file. For example,

```
&IF ^Q1 &THEN &G LABEL1
```

If ^Q1 contains true, control is transferred to the command line that follows the line labeled LINE1. If ^Q1 contains false, control is transferred to the next command line after this one.

&L: Label function

Defines a labeled location that can be the object of an &G (label) function. For example,

```
&IF ^Q1 &THEN &G LABEL2
```

```
&
```

```
PR $F2>VOL2>JOHN.XX
```

```
&
```

```
&L LABEL2
```

```
DL $F2>VOL2>JOHN.XX
```

```
&
```

If ^Q1 contains “true,” JOHN.XX is deleted. Otherwise, the content of JOHN.XX is displayed by the PR command line.

&Q: Quit command-file execution

Stops execution of the command file and returns control to the Command Processor display. For example,

```
&Q4 "ARE YOU READY TO QUIT?"
```

```
&
```

```
&IF ^Q4 &THEN &Q
```

```
&
```

```
PR $F2>VOL2>JOHN.XX
```

If the user keys in Y in response to the question, command-file execution stops. If the user keys in N, execution continues with the PR command line.

R(EAD): Read function

Reads one line from the text file that was opened in the FILE directive. Variables are substituted (^R1 to ^R32) for as many strings as are on the line. The READ function must be used in conjunction with the &IF directive. For example,

```
&IF NOT READ &THEN &G bye
```

AND: AND function

Returns a value of true if the values in both (active function) values are true. The AND function can be an active function for another active function, in which case, it is enclosed in parentheses. For example,

```
&IF ^Q1 AND ^Q2 &THEN &G LABEL1
```

```
&
```

```
&IF ^Q5 AND (^Q1 AND ^Q2) &THEN &G LABEL5
```

```
&
```

```
PR $F2>VOL2>JOHN.XX
```

For the first command line, if the values in both ^Q1 and ^Q2 are true, control is transferred to the first command line after the line designated LABEL1. For the second command line, if all of the values in ^Q5, ^Q1, and ^Q2 are true, control is transferred to the first command line after the line designated LABEL5 and if any of these values is false, execution continues with the PR line. Any unnecessary pairs of parentheses are ignored if the syntax is otherwise correct.

OR: OR function

Returns a value of true if the values in either of the (active function) values are true. The OR function can be an active function for another active function, in which case, it is enclosed in parentheses. For example,

```
&IF ^Q1 OR ^Q2 &THEN &G LABEL1  
  
&  
  
&IF ^Q5 OR (^Q1 AND ^Q2) &THEN &G LABEL5  
  
&  
  
PR $F2>VOL2>JOHN.XX
```

For the first command line, if either ^Q1 or ^Q2, or both, are true, control is transferred to the first command line after the line designated LABEL1. For the second command line, if either ^Q5 or (^Q1 AND ^Q2), or both, are true, control is transferred to the first command line after the line designated LABEL5 and if ^Q5 is false and (^Q1 AND ^Q2) is false, execution continues with the PR line. Any unnecessary pairs of parentheses are ignored if the syntax is otherwise correct.

NOT: Invert function

If (active function) contains true, “false” is returned. If (active function) contains false, “true” is returned. For example,

```
&IF NOT EXISTS NET>&LDR>BOOTNAME.BO &THEN &G LABEL1  
  
&
```

If the file named by NET>&LDR>BOOTNAME.BO does not exist, control is transferred to the next command line after the location designated LABEL1, but if it does exist, execution continues with the next command line after this one.

EXIST: Does this volume/directory or file exist?

If (pathname) defines a volume, directory, or file that is found, a “true” is returned. For example,

```
&IF EXISTS NET>&LDR>BOOTNAME.BO &THEN &G LABEL1  
  
&  
  
&  
  
&IF EXISTS $F1>&LDR> &THEN &G LABEL5
```

For the first command line, if file BOOTNAME.BO is found in directory &LDR on an HM, the result is true, and control is transferred to the next command line after the line labeled LABEL1. For the second command line, if a volume or directory named &LDR is found on the removable media in drive 1, the result is true, and control is transferred to the next command line after the line labeled LABEL5. If a pathname doesn't define an existing volume, directory, or file, the result is false and the next command line is executed, rather than the one pointed to by the &G directive. Wildcard characters can be used in the pathnames.

Reference: Section 11.18, Short Cuts and “Wildcard” Characters (Miscellaneous Commands).

^Rn: READ variables

READ variables are substituted for strings on a line. They number from 1 through 32 and are used with the READ directive which reads one line from an open text file. For example,

```
&P “Requested by EC NET>TEST>FILES.EC NET TEST FILES XX”  
  
&FILE &1>&2>&3.&4  
  
& L NEXT  
  
&IF NOT READ &THEN &G DONE  
  
&P NAME ^R1>^R2>^R3.^R4  
  
PR ^R1>^R2>^R3.^R4  
  
&G NEXT  
  
&L DONE  
  
&P END
```

^Qn: Yes or No Answer

This item returns a value of true or false, according to the last answer provided to the corresponding &Qn question. The value is true when the answer was Y (yes) and it is false when the answer was N (no) (press <ENTER> after keying Y or N). If the question has not yet been answered, an error message is generated. For example,

```
&IF ^Q1 DL NET>VOL1>JOHN.XX  
&  
DL NET>VOL1>JANE.XX
```

If the answer to question &Q1 was “yes,” file JOHN.XX is deleted, and if the answer was “no,” file JANE.XX is deleted.

^Vn: Character- string answer

This item returns the character string keyed-in response to the corresponding &Vn question. If the question has not yet been answered, an error message is generated. For example,

```
&V1 "WHERE DO YOU WANT THIS FILE TO BE STORED?"  
&  
CP NET>VOL1>MYFILE.XX ^V1= -D  
&
```

When the question appears on the screen, execution pauses to wait for the answer. The user keys in pathname NET>VOL1> as the answer, and that answer is stored in ^V1. When the copy command line is executed, the content of the ^V1 item (NET>VOL1>) is used as the pathname for the destination of the file to be copied.

&S: Command Store

This item stores a 1- to 32-character string in ^V1 or stores a “Y” or an “N” in ^Q1.

```
&S &V1 string
```

or

```
&S Y &Q1
```

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